Quantum Gravity as Torsional Mismatch in Biquaternion Double Spacetime

https://github.com/hypernumbernet

November 8, 2026

Abstract

General relativity describes gravity as the curvature of a single spacetime continuum, yet its reconciliation with quantum mechanics remains open: singularities, the continuum hypothesis, and the missing microscopic origin of curvature stand in the way. This paper develops the quantum sector of the Double Spacetime Theory (DST) introduced in the companion work. There spacetime is not a shared manifold but a pair of compactified Minkowski spacetimes attached to every massive particle and encoded in the 16-real-dimensional biquaternion algebra $\mathbb{H} \oplus \mathbb{H} \cong \mathrm{Cl}(3,1)$. Gravity is the torsional mismatch J between the usual and the dual rotor, and on admissible configurations $|J| \leq 3\pi^2/8$.

The dual generators act on $\mathbb{C}^2$ by $-i\sigma_a$, and the dual rotors are exactly the matrices of SU(2). A unit dual spinor is a point of the unit sphere; a dual rotor turns its spin vector by the full angle, the interference of u with $U(\beta)u$ has intensity $\cos^2(\|\beta\|/4)$, and on that sphere the Robertson–Schrödinger relation for two spin axes is an equality. The four-vector whose time component is $\|u\|^2$ and whose spatial part is the spin vector lies on the future light cone, every future lightlike four-vector arises in this way, and a phase of unit modulus leaves that four-vector fixed. A superposition does not appear as the sum of the two four-vectors. The Dirac matrices are the grade-1 generators of $\mathrm{Cl}(3,1)$, and the momentum-space Dirac equation is the first-order factor of the Minkowski quadratic. On the mass shell every dual spinor u determines a solution, the pair (u, iu) at rest. The equation is covariant under every total rotor, and every total rotor is uniquely a usual rotor followed by a dual rotor. The dual rotors are exactly the total rotors that fix the rest momentum, so the dual sector is the little group of a massive particle; usual rotors along x and z compose to a usual rotor followed by a dual rotation about y, the Thomas–Wigner rotation. Under $p_\mu \mapsto -i\partial_\mu$ the equation becomes the spacetime Dirac equation. No operators on the two-dimensional dual sector satisfy $[A, B] = iI$. On differentiable fields over the chart, $X^\nu \Psi(x) = x^\nu \Psi(x)$ and $P_\mu = -i\partial_\mu$ satisfy $[X^\nu, P_\mu] = i\delta^\nu_\mu$, the only nonzero bracket of the chart pair. A plane wave is an eigenfield of each momentum component and of no position component, and the only joint eigenfield of position and momentum is zero. Its plane waves carry the de Broglie phase, its current is conserved, and the time component of that current is the nonnegative density $\Psi^\dagger \Psi$ that Born's rule requires; the Killing self-pairing is indefinite and is not that density. Minimal coupling to an electromagnetic potential is gauge covariant. In the resulting second-order equation the magnetic field acts through the dual generators, identically on both Weyl blocks and with gyromagnetic ratio 2, and the electric field through the usual generators. Spin precession in a magnetic field is a dual rotation. On Dirac spinors the dual map acts by $i\gamma^5$ and reverses the four-momentum. The dual spinor carries an antilinear quaternionic structure with square -1 that commutes with the dual rotors and reverses the spin, and the charge conjugation it defines reverses the charge and the momentum. Their composite, taken with the point reflection of spacetime, carries every solution to a solution of opposite charge in the reflected potential and keeps the momentum of every plane wave: the antiparticle is the charge conjugate read through the time reversal of the dual map. Two dual spinors form a joint state in the tensor product, and no local dual rotor changes its entanglement. The joint states fixed by every common dual rotor are the multiples of the singlet, which pairs every dual spinor with its quaternionic partner. Exchange splits the joint space into that singlet, of total spin 0, and the symmetric states, of total spin 1; two dual spinors on one ray cannot occupy the singlet. The orthogonal complement of that ray is the spin-reversed partner, and the two rays exhaust the space. The singlet correlation along two axes is $-a \cdot b$; no local operation on one particle changes the statistics of the other, and the CHSH combination reaches $2\sqrt{2}$, beyond the bound 2 that holds whenever each outcome is fixed by variables of its own particle.

The algebra does not fix the dynamics. Equal-scale interference nodes run inward and do not reproduce the Bohr radii. One revolution of the dual rotor restores every point of its orbit and leaves the rotor at -1, at every radius. The plane-wave factor on that revolution equals 1 exactly when $pr = n\hbar$, and the circular path then lies at $n^2 a_0 \gamma_s$, strictly inside the Coulomb radius by a fraction less than 1/60000. The written opposite-sign particle action frees the mismatch and sources the common

rapidity; its Jacobi integral is indefinite, and the harmonic stiffness neither restores a frequency nor selects a misalignment. The value of the mass, measurement, and the path integral remain geometric hypotheses, and the internal duality of the algebra is not ten-dimensional spacetime supersymmetry.

1 Introduction: An Algebraic Approach to Quantum Gravity

The reconciliation of general relativity with quantum mechanics remains one of the most profound open problems in theoretical physics. Despite decades of effort, no consensus framework has emerged that is simultaneously consistent with the empirical successes of both theories while resolving their conceptual incompatibilities. Loop quantum gravity discretizes geometry but retains a background manifold and curvature as fundamental; string theory introduces extra dimensions and supersymmetry without direct observational support; asymptotic safety and causal dynamical triangulations struggle with the continuum limit and the origin of spacetime itself. All existing approaches share a common foundational assumption: the spacetime continuum hypothesis. Spacetime is presumed to exist as a smooth, differentiable manifold independent of matter, with gravity arising from its curvature sourced by energy-momentum.

This assumption, while extraordinarily successful at macroscopic scales, leads to severe difficulties at the quantum level: ultraviolet divergences on curved backgrounds, the black hole information paradox, cosmological singularities, and—most fundamentally—the absence of any physical mechanism explaining why or how matter should curve spacetime. Mach's principle remains unfulfilled by that continuum. Once the continuum is refused, other matter fixes the mismatch a particle expresses, and does not fix the stiffness with which that mismatch accelerates the common rapidity.

In this paper we develop a different approach. The underlying algebra is the 16-real-dimensional biquaternion algebra isomorphic to $\mathrm{Cl}(3,1)$; a background manifold and Riemann curvature are not required for the identities recorded below. The physical picture is the pair of rotors R_{usual} and R_{dual}. Dual-sector kinematics is the representation on $\mathbb{C}^2$; gravitational phenomenology is torsional mismatch measured by $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$. Identification of J with the teleparallel scalar, and Einstein–Hilbert equivalence, are taken from the TEGR literature after a rotor-to-tetrad dictionary is fixed.

The dual rotor at every rapidity is the Rodrigues matrix of SU(2), and every matrix of SU(2) arises in this way (Section 3). A unit dual spinor is a point of the unit sphere, a dual rotor turns its spin vector by the full angle, and the Robertson–Schrödinger relation for two spin axes is an equality on that sphere. The Dirac matrices of signature $(-,+,+,+)$ are the grade-1 generators of $\mathrm{Cl}(3,1)$; they obey $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ and are not the hyperbolic bivectors used for P_{spin}. The momentum-space Dirac equation follows from the Clifford square of the Minkowski quadratic (Section 4). It is covariant under every total rotor, and the dual factor of a total rotor is the little group of the rest momentum, so spin is carried by the dual sector and momentum by the usual sector. The body of the information is a vector of that dual sector, and the usual chart receives exactly the future light cone of Section 3.1. Replacing p_μ by $-i\partial_\mu$ gives the Dirac equation on spacetime. Its conserved current has the nonnegative density that Born's rule requires, whereas the Killing self-pairing has signature $(3,3)$ and is not a probability [1]. Minimal coupling to an electromagnetic potential is gauge covariant. In the coupled equation the magnetic field acts on the spinor through the dual generators, so the magnetic moment belongs to the dual sector and spin precession in a magnetic field is a dual rotation. The dual map acts on Dirac spinors by $i\gamma^5$ and reverses the four-momentum; composed with the charge conjugation supplied by the quaternionic structure of the dual spinor, it produces the antiparticle. For two particles the same quaternionic structure builds the singlet, the unique joint state that common dual rotation leaves fixed. Exchange splits the joint space into that singlet and the symmetric states of total spin 1, and two dual spinors on one ray cannot occupy the singlet. The orthogonal complement of that ray is the spin-reversed partner, and the two rays exhaust the space. The correlations carry no signal, and they exceed every assignment of outcomes by particle-local variables, so entanglement is not a correlation of the dual phases of the two particles. Position as an operator, and mass as torsion, are not derived here. The phase of a plane wave separates a sharp momentum from a chart-dependent reading.

Abandoning the continuum hypothesis also bounds the singularities of classical general relativity. The compactness of the dual spacetime discretizes the rotor angles, and on admissible configurations the algebraic ceiling $|J| \le 3\pi^2/8$ (equivalently $|J_{\mathrm{norm}}| \le 1$) forbids the unbounded curvature that would otherwise appear at $r = 0$ or at the Big Bang, provided the configuration remains admissible. In the same picture event horizons are replaced by finite torsional layers that light traverses by resonance tunneling, and strong-field deviations from general relativity become observable in principle. Dual-only kinematics on the admissible cone determine when J can be cancelled or reversed and allow the dual budget to be routed across silent axes. The pure-boost point $|J_{\mathrm{norm}}| = 1$ admits no other dual completion. That point

meets the unsigned-mass ceiling, and so do seven further corners, at the normalised heights 1, $\frac{1}{3}$, $-\frac{1}{3}$, and -1; balance reaches at most five sixths of the same ceiling. Whether laboratory fields supply that dual force is the experimental question of Section 12.

Section 2 constructs the square-$+1$ projectors of the algebra. Section 3 represents the dual sector on $\mathbb{C}^2$, separates the Killing pairing from Born's rule, and records the spin vector, the 4π interference, the spin uncertainty equality, and the lightlike four-vector by which a dual spinor enters the usual chart. Section 4 derives the Dirac equation from the Clifford square: its on-shell solutions, the mass bilinear, the current and the spin along the momentum, the covariance under total rotors with the little group and the Thomas–Wigner rotation, the Gordon identity, the spacetime equation with its Born density and the separation there of a sharp momentum from a chart-dependent reading, the coupling to an electromagnetic potential, and the antiparticle obtained from the dual map and charge conjugation. Section 5 treats two dual spinors: the singlet, the two rays of one dual spinor, exchange and the spin-0/spin-1 split, its correlations, the absence of signalling, and the CHSH bound. Section 6 compares equal-scale interference nodes with the Bohr radii and places the hydrogen series on the restoring side of the outer well. Sections 7 and 8 treat the torsional stiffness and the written particle action, and Section 9 reads the opposite signatures of the two sectors as an intrinsic supersymmetry. Section 10 presents conjectural predictions bounded by the admissible ceiling, Section 11 compares the framework with other programmes, and Section 12 outlines near-term tests. Section 13 gathers the algebraic picture, and Section 14 summarizes.

Quantum gravity is not claimed to follow in full from $\mathrm{Cl}(3,1)$. The results of this paper are algebraic identities in the 16-dimensional algebra; continuum dynamics remains a classical approximation.

2 Biquaternion Spinors and the Minimal Left Ideal

The biquaternion algebra $\mathbb{H} \oplus \mathbb{H}$, canonically isomorphic to the Clifford algebra $\mathrm{Cl}(3,1)$ of Minkowski spacetime with signature $(-,+,+,+)$, provides a natural and minimal realization of Dirac spinors. Unlike conventional approaches that introduce spinors as an external representation space, here the spinor emerges directly as an element of the algebra itself.

A primitive idempotent that projects onto an irreducible representation must be built from generators that square to $+1$. If g satisfies $g^2 = +1$, then

$$\frac{1+g}{2}$$

is idempotent. Write $j := e_0 e_1$ for the hyperbolic generator of axis 0 (so $j^2 = +1$). This j is *not* the time vector $\gamma^0 = e_0$, which squares to -1 in signature $(-,+,+,+)$ (Section 4). The working projectors are

$$P_{\mathrm{spin}} = \frac{1+j}{2} = \frac{1+e_0 e_1}{2}$$

and the complementary spatial chirality pair

$$P_L = \frac{1-e_1}{2}, \qquad P_R = \frac{1+e_1}{2}$$

(with $e_1^2 = +1$). They satisfy $P_L^2 = P_L$, $P_R^2 = P_R$, and $P_L + P_R = 1$. The pair P_L, P_R organizes a chiral split in the algebra; it does not by itself yield the weak interaction of the Standard Model.

The product

$$P = P_{\mathrm{spin}}\, P_R$$

is not idempotent: $e_0 e_1$ and e_1 anticommute, the factors do not commute, and $(P_{\mathrm{spin}} P_R)^2 \neq P_{\mathrm{spin}} P_R$. A commuting square-$+1$ pair is e_1 with the off-axis boost $e_0 e_2$; their product projector is idempotent and generates a left ideal of real dimension eight. That ideal is the direct sum of the four-dimensional Minkowski core $\mathrm{Cl}(3,1)\,P$ and a copy along the null generator e_4; the copy is a proper $G(3,1,1)$-submodule, so the eight-dimensional ideal is reducible as a PGA-module. The match with the real dimension of a Dirac spinor $\mathbb{C}^4$ is this null doubling, not an identification with an irreducible Dirac module. Irreducibility of the Minkowski core as a $\mathrm{Cl}(3,1)$-module is not claimed.

By contrast, the naive formulae built from the $\mathrm{Cl}(3,1)$ pseudoscalar i alone,

$$\frac{1 \pm i}{2} \qquad (i^2 = -1),$$

are not idempotent: $\big((1-i)/2\big)^2 = -i/2$. Composites such as $i\gamma_1\gamma_2$ that square to $+1$ remain admissible building blocks; the bare pseudoscalar does not.

A general element of a candidate spinor space takes the explicit form

$$\psi = \big(a_0 + a_1 i + b_1 kI + b_2 kJ + b_3 kK + c_1 jI + c_2 jJ + c_3 jK\big)P,$$

where $a_0, a_1, b_i, c_i \in \mathbb{R}$ and P is a square-$+1$ projector (not $P_{\text{spin}}P_R$). Identifying the pseudoscalar i with the imaginary unit of quantum mechanics is a dictionary, not a proof that the carrier is the Dirac module. The dual map $X \mapsto Xi$ still reverses time orientation, and on Dirac spinors it reverses the four-momentum (Section 4.8); the algebraic projectors themselves are P_L and P_R above, not $(1 \pm i)/2$.

The action of rotors on algebra elements is left multiplication. Same-axis usual and dual generators commute; off-axis pairs need not. In particular Γ_a does not commute with every usual-sector generator $i\Gamma_b$, so the order of the factors in $R_{\text{total}} = R_{\text{usual}}R_{\text{dual}}$ matters off-axis. With that order fixed the factorization is an identity in the $\mathbb{C}^2$ representation: every total rotor is a usual rotor followed by a dual rotor, and both factors are unique (Section 4).

The bilinear $\bar{\psi}\psi \propto \text{Tr}(\Omega_{\text{biv}})$ and the emergence of the full fermionic Standard Model from the 16-dimensional algebra are not established here. What follows from the algebra is the projector calculus above.

3 Quantum States as Dual-Spacetime Rotations

The dual rotor

$$R_{\text{dual}} = \exp\left(\sum_{a=1}^{3} \frac{\phi_a}{2}\Gamma_a\right),$$

where $\Gamma_1 = I$, $\Gamma_2 = J$, $\Gamma_3 = K$ satisfy the quaternion relations $\Gamma_a^2 = -1$ and $IJ = K$, is represented on $\mathbb{C}^2$ by $-i\sigma_a$. The alternative dictionary $I \leftrightarrow i\sigma$ is incompatible with $IJ = K$.

At every dual rapidity β the exponential is the Rodrigues formula

$$U(\beta) = \cos(\|\beta\|/2)\, I + \sin(\|\beta\|/2)\, n \cdot (-i\sigma),$$

with $n = \beta/\|\beta\|$ when $\beta \neq 0$ and $U(0) = I$. That matrix is unitary of determinant 1, and conversely every unitary matrix of determinant 1 is a dual rotor: the dual rotors are exactly the group SU(2). The inverse is the opposite rapidity, so a relative dual rotor is a product in the group: its character records the alignment of the two dual axes, and the noncommuting residue is a traceless tilt. A 2π dual rotation about a computational axis is $-I$, and along a fixed axis the dual angle is periodic modulo 4π, as the compactness of the dual spacetime requires. That period is visible in interference: the intensity of $u + U(\beta)u$ is $4\cos^2(\|\beta\|/4)\,\|u\|^2$ for every dual spinor, so a 2π dual rotation, which sends u to $-u$, is not the identity of the superposition.

A unit dual spinor is a point of the unit sphere. Its spin vector $r_k = \langle u, \sigma_k u\rangle$ is real, satisfies $\langle u, \sigma \cdot a\, u\rangle = a \cdot r$, and has $|r| = 1$. The outcome $s = \pm 1$ along a unit axis a occurs with probability $(1 + s\, a \cdot r)/2$. A dual rotor $U(\beta)$ turns r by the full angle $\|\beta\|$ about β, and two matrices of SU(2) turn every axis alike exactly when they differ by sign.

The dual generators supply the spin commutators $[\sigma \cdot a, \sigma \cdot b] = 2i\,\sigma \cdot (a \times b)$. On a unit dual spinor the Robertson–Schrödinger relation is an equality: $\text{Var}(\sigma \cdot a)\,\text{Var}(\sigma \cdot b)$ equals $(a \cdot b - \langle \sigma \cdot a\rangle\langle \sigma \cdot b\rangle)^2 + \langle \sigma \cdot (a \times b)\rangle^2$, so the bound $|\langle[\sigma \cdot a, \sigma \cdot b]\rangle|^2/4 \leq \text{Var}(\sigma \cdot a)\,\text{Var}(\sigma \cdot b)$ holds with the covariance as the exact excess. Along three perpendicular unit axes the variances sum to 2. Every commutator on this two-dimensional space is traceless, and the trace of iI is $2i$, so no operators on the dual sector obey $[A, B] = iI$. The spin commutators live there because they are traceless. Position and momentum are operators on differentiable fields over the chart: X^ν multiplies by the coordinate x^ν, and $P_\mu = -i\partial_\mu$. On every such field $[X^\nu, P_\mu] = i\delta^\nu_\mu$. The coordinates commute, and on a twice differentiable field so do the momenta. A sharp momentum and a chart-dependent reading are already separated by the de Broglie phase of Section 4. An uncertainty bound $\Delta\phi \cdot \Delta\theta \geq 1/2$ for orbital angles remains open. An amplitude on the discrete dual circle $\mathbb{Z}/N\mathbb{Z}$ is a different finite Hilbert space. Its Fourier transform onto winding labels inverts, and $\sum_k |\hat{f}(k)|^2 = N \sum_j |f(j)|^2$, so the normalised transform is unitary; on the three dual angles the factor is N^3. That space is not $\mathbb{C}^2$, and the transform is not the pair $[X^\nu, P_\mu]$. What the circle does carry is a pair of finite order. Shifting the angle by s and multiplying by the character $\exp(-2\pi i\, rj/N)$ both preserve the inner product; each returns the amplitude after N steps and after no smaller positive number of steps; and shift followed by the character is $\exp(-2\pi i\, rs/N)$ times the reverse order. The

transform turns a shift into multiplication of winding k by $\exp(-2\pi i\, ks/N)$, and turns multiplication by the character into the reading of the original transform at label $k+r$. Circular convolution becomes the product of the transforms. For a nonzero amplitude the numbers A and B of occupied sites and occupied windings satisfy $AB \geq N$, with equality for one occupied site and for one occupied winding. On the three dual angles the bound is N^3, and an amplitude that separates across the axes has transform equal to the product of the three circle transforms.

The probabilistic interpretation is not the Killing self-pairing. The identification

$$|\psi|^2 \propto \tfrac{1}{2}B(\Omega_{\mathrm{biv}}, \Omega_{\mathrm{biv}})$$

fails: the Killing pairing has signature $(3,3)$ and is negative on pure elliptic configurations. Self-overlap is therefore not a probability. Born's rule belongs to the Hilbert layer and is not a consequence of J [1]. The nonnegative conserved density it requires is the time component of the Dirac current, $\Psi^\dagger\Psi$ (Section 4).

Measurement and collapse as torsional synchronization are likewise left open; the kinematics of two entangled dual spinors is fixed in Section 5. Dual-sector kinematics in this paper means the $\mathbb{C}^2$ representation of the cyclic generators.

3.1 The light cone as the filter

The state does not live in the joint rapidity space. The Killing pairing on that space has signature $(3,3)$ and is negative on a pure elliptic configuration, so it is not an inner product, and its self-overlap is not a probability. The body of the information is a vector u of the dual sector $\mathbb{C}^2$, the space on which the dual rotors act by unitary matrices of determinant one. Two such vectors carry the same information when one is the other multiplied by a complex number of modulus one. That phase cancels in the squared norm and in every spin expectation.

What the usual chart receives is the four-vector $m(u)$ whose time component is the information weight $\|u\|^2$ and whose spatial part is the spin vector $r(u)$, with $r_k = \langle u, \sigma_k u\rangle$. The identity $|r(u)|^2 = \|u\|^4$, read in the Minkowski quadratic $Q = -t^2 + x^2 + y^2 + z^2$ of signature $(-,+,+,+)$, is $Q\big(m(u)\big) = 0$. The time component is nonnegative, and $m(u)$ vanishes if and only if u does. On a unit dual spinor the time component is 1 and the spatial part is a unit vector, so the image is a future null ray in the direction of the spin. A phase of unit modulus leaves $m(u)$ fixed. A general complex rescaling multiplies $m(u)$ by the squared modulus, and the image stays on the same null ray. A dual rotor preserves the time component and turns the spatial part by the full angle about its axis; the image remains lightlike. No timelike vector and no spacelike vector is written by this map. Conversely, every four-vector that is lightlike and has nonnegative time component is $m(u)$ for some dual vector u. The image of the filter is the future light cone, the origin included.

Superposition is formed before the filter, and it is not addition in the chart. The appearances of the computational basis are $(1,0,0,1)$ and $(1,0,0,-1)$. Their sum is the timelike vector $(2,0,0,0)$. The appearance of their superposition is the null vector $(2,2,0,0)$, and the discrepancy $(0,2,0,0)$ is spacelike. A phase of unit modulus on the second vector alone leaves that vector's appearance fixed, while the same phase, taken as a relative phase, sends the superposition to the different null vector $(2,0,2,0)$. The two superpositions do not carry the same information. What remains of addition is the parallelogram identity $m(u+v) + m(u-v) = 2m(u) + 2m(v)$: the cross term cancels between the two relative signs and is not itself an appearance. Interference therefore lives in the dual sector. The chart receives only the lightlike result.

Information is at home in the compact dual sector. It becomes readable in the usual sector only as this future lightlike four-vector. The constancy of the speed of light is that identity: the spin cannot outrun the weight. The indefinite pairing of the two sectors together remains the mismatch J. It is not a second copy of the state, and the whole of $\mathrm{Cl}(3,1)$ is not assembled here into one Hilbert space. The canonical pair is not obtained from the filter, nor is a quantization of the mismatch dynamics: the dual space cannot carry $[A,B] = iI$. The chart point enters through the phase of a plane wave, and a sharp momentum there excludes a position-dependent reading. The operators that answer that phase act on fields over the chart, as in Section 4.

A Dirac spinor carries two such dual vectors, the upper Weyl block u and the lower block l. The time component of the probability current is the sum of the two information weights, $\|u\|^2 + \|l\|^2$, and its spatial part is the lower spin vector minus the upper one. The current is therefore the lower image $m(l)$ plus the upper image with its space reversed. When only one chirality is present the current stays lightlike: the lower block's current is $m(l)$ itself, and the upper block's current is $m(u)$ with its space reversed. The massless Weyl equation is solved on the upper block by placing the null momentum at

$m(u)$, and on the lower block by the same image with its space reversed. In this representation the spatial part of the current and the spatial part of the momentum point opposite ways, the same sign as the longitudinal on-shell current along $(\cosh\alpha, 0, 0, -\sinh\alpha)$.

The rest pair is (u, iu). The phase i has modulus one, so the two chiral images are the same null vector $m(u)$. The current reads one of them with its space reversed. The spatial parts cancel and the weights add, and the current is $(2\|u\|^2, 0, 0, 0)$. Whenever u does not vanish this is timelike, with Minkowski square $-(2\|u\|^2)^2$, while the sum of the two images is still lightlike. Rest mass is the coupling that hands the current two chiral copies of one dual vector as null images that are spatial mirrors of each other. The filter never writes a timelike vector from one dual vector. A timelike current is the signed sum of two lightlike images. When the mass vanishes the blocks separate and each current stays on the cone. When the mass does not vanish they mix, and on the forward mass shell the assembled current is timelike.

4 The Dirac Equation from the Clifford Square

4.1 Clifford relations and the momentum-space equation

The Dirac matrices of signature $(-,+,+,+)$ are the grade-1 generators γ^μ of $\mathrm{Cl}(3,1)$. They obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \qquad \eta = \mathrm{diag}(-1,+1,+1,+1).$$

In particular $(\gamma^0)^2 = -1$. The hyperbolic generator $j = e_0e_1$ used for P_{spin} squares to $+1$, so $\gamma^0 \neq j$, and the dictionary $\gamma^0 \leftrightarrow j$, $\gamma^i \leftrightarrow kI, \ldots$ is replaced by this grade-1 basis.

For a four-momentum p the grade-1 element $\gamma(p) = \sum_\mu p_\mu\gamma^\mu$ squares to the Minkowski quadratic $Q(p) = -p_0^2 + p_1^2 + p_2^2 + p_3^2$, both in $\mathrm{Cl}(3,1)$ and as a matrix on $\mathbb{C}^4$. The two chiral slashes of the Weyl representation multiply to the same quadratic. The mass shell $Q(p) = -m^2$ is therefore the vanishing of $(\gamma(p) - im)(\gamma(p) + im)$, and a nonzero spinor with

$$\gamma(p)\Psi = im\Psi$$

lies on that shell: this is the Dirac equation in momentum space. In the Weyl representation $\gamma(p)$ is off-diagonal, so vanishing mass decouples the two chiral blocks while a nonzero mass mixes them. Chirality γ^5 sends a solution of mass m to a solution of mass $-m$.

4.2 On-shell solutions and the longitudinal boost

On the mass shell with $m \neq 0$ an arbitrary dual spinor u determines a solution: the upper Weyl block is u, and the lower block is fixed by the chiral slash of p. At rest momentum $(m, 0, 0, 0)$ this is the pair $\Psi = (u, iu)$. Dual rotation of u remains a solution at the same four-momentum: the dual rotor acts internally on the Weyl parameter.

A longitudinal boost of rest momentum,

$$p = (m\cosh\alpha,\ 0,\ 0,\ m\sinh\alpha),$$

stays on the shell, and at that momentum the Weyl construction is the Dirac-boosted rest pair. The boost acts by the hyperbolic Rodrigues factor

$$V(\alpha) = \cosh(\alpha/2)\, I + \sinh(\alpha/2)\, \sigma_z$$

on the upper Weyl block and by $V(-\alpha)$ on the lower: the rest pair (u, iu) is sent to the on-shell solution whose Weyl parameter is $V(\alpha)u$. Dual rotation about the boost axis commutes with $V(\alpha)$, so internal dual rotation and the longitudinal boost may be taken in either order. Dual rotation about a perpendicular axis does not commute with $V(\alpha)$.

4.3 Overlap, mass bilinear, and current

The $\mathbb{C}^4$ overlap of two rest-frame solutions that differ by a dual rotation is twice the dual-rotor amplitude. A longitudinal boost of rapidity α multiplies that overlap by $\cosh\alpha$, for every dual rotation, so the overlap agrees with its rest-frame value only at vanishing rapidity. It is a Euclidean identity on $\mathbb{C}^4$, not a spacetime inner product.

The invariant pairing is the mass bilinear $\bar\Psi\Phi = \Psi^\dagger(-i\gamma^0)\Phi$. On two rest-frame solutions it coincides with the Euclidean overlap, and therefore with twice the dual-spinor pairing, and a longitudinal boost leaves it unchanged. The Dirac current built from the same matrix is

$$j^\mu = \Psi^\dagger(-i\gamma^0)\gamma^\mu\Phi = i\,\bar\Psi\Phi\,(\cosh\alpha,\,0,\,0,\,-\sinh\alpha)$$

along the boosted pair. In this representation $(-i\gamma^0)\gamma^0 = i$, so the time component is i times the Euclidean overlap and carries the factor $\cosh\alpha$. The transverse components vanish. The Clifford contraction of the current with the boosted momentum equals $im\bar\Psi\Phi$, and $mj^0 = i\bar\Psi\Phi\,p^0$, $-mj^3 = i\bar\Psi\Phi\,p^3$, so for $m \neq 0$ the combination $(j^0, -j^3)$ is parallel to p.

The eigenrays of σ_z label the spin of this solution along the momentum. $V(\alpha)$ multiplies the eigenvalue $+1$ by $e^{\alpha/2}$ on the upper block and by $e^{-\alpha/2}$ on the lower, and both blocks remain in that ray. The eigenvalue -1 exchanges the two factors, so at large positive rapidity the first spin lies mostly in the upper block and the second mostly in the lower. The mass bilinear between the two rays vanishes at every longitudinal rapidity, and so does every component of the current between them. A dual rotation by θ about the boost axis multiplies the eigenvalue $+1$ by $e^{-i\theta/2}$ and the eigenvalue -1 by $e^{i\theta/2}$; a dual rotation of angle π about a perpendicular axis exchanges the two rays.

4.4 Total-rotor covariance and the little group

The longitudinal boost is one instance of a covariance under the whole total-rotor group. In the $\mathbb{C}^2$ representation a total rotor is a complex matrix M of determinant one. A four-momentum is the Hermitian matrix $\hat p = p_0 I + \sum_a p_a\sigma_a$, whose determinant is $-Q(p)$, and M acts on it by $\hat p \mapsto M\hat pM^\dagger$; this defines the Lorentz action $\Lambda(M)$ on momenta. On a Dirac spinor M acts by $S(M)$: by M on the upper Weyl block and by $(M^\dagger)^{-1}$ on the lower. The action preserves $Q(p)$ and intertwines both chiral slashes, so it sends every solution of mass m at p to a solution of the same mass at $\Lambda(M)p$. The mass bilinear is invariant, and the Clifford contraction of the current with a four-vector is unchanged when the vector is transformed together with the spinors. A dual rotor is unitary and acts by the same matrix on both blocks. A usual rotor $V(\alpha) = \exp(\alpha\cdot\sigma/2)$ is Hermitian and acts on the lower block by its inverse, as $V(-\alpha)$ does for the longitudinal boost.

Every total rotor is a usual rotor followed by a dual rotor, $M = V(\alpha)U(\beta)$, and both factors are unique: $V(\alpha)$ is the positive square root of $MM^\dagger$, and $U(\beta) = V(\alpha)^{-1}M$. A total rotor fixes the rest momentum $(m,0,0,0)$ with $m \neq 0$ exactly when it is a dual rotor. The dual sector is therefore the little group of a massive particle, and its spin degrees of freedom are those of the dual spinor u. The momentum is carried by the usual factor: every momentum on the forward mass shell is reached from rest by exactly one usual rotor. Usual rotors along perpendicular axes do not commute, and their product is not a usual rotor: the product of usual rotors along x and z is a usual rotor followed by a nontrivial dual rotation about y. This dual factor is the Thomas–Wigner rotation. In double spacetime it is an internal rotation of the dual spinor, produced by composing usual rotors. The same split organises a Fermat triple. The boost and the cyclic angle lie on different axes: degree two is a pure dual rotor, and every positive solution of degree at least three is mixed.

4.5 Gordon identity and the Born density

The current of two solutions at one momentum is fixed by that momentum. The adjoint matrix $-i\gamma^0$ is Hermitian and every $(-i\gamma^0)\gamma^\mu$ is anti-Hermitian, so the anticommutator $\{\gamma(p),\gamma^\mu\} = 2p^\mu$ yields the Gordon identity

$$m\,j^\mu = -i\,p^\mu\,\bar\Psi\Phi, \qquad p^\mu = \eta^{\mu\nu}p_\nu,$$

for any two solutions Ψ and Φ of mass m at p. The longitudinal current above is its instance along z. The time component reads $m\,\Psi^\dagger\Phi = p_0\,\bar\Psi\Phi$: at every on-shell momentum the Euclidean overlap is p_0/m times the invariant bilinear, and $\cosh\alpha$ is that ratio for the longitudinal boost.

For a single spinor the current is i times a real four-vector J^μ with $J^0 = \Psi^\dagger\Psi$. On a nonzero solution with $m \neq 0$ the component p_0 does not vanish, $\bar\Psi\Psi = (m/p_0)\,\Psi^\dagger\Psi$ is real and nonzero, and J has Minkowski square $-(\bar\Psi\Psi)^2$. The real current is therefore timelike, and its spatial part points opposite the spatial momentum; its invariant length is the mass bilinear. On the rest pair the two chiral images are the same null vector, and the real current falls to the pure time vector $(2\|u\|^2,0,0,0)$. With $m > 0$ and $p_0 > 0$ the mass bilinear is positive and does not exceed $\Psi^\dagger\Psi$. A total rotor that carries the rest momentum to p multiplies the density $\Psi^\dagger\Psi$ of a rest solution by p_0/m, while every dual rotor leaves it unchanged. The density is fixed by the little group and changes only with the usual factor.

4.6 The spacetime equation

Replacing p_μ by $-i\partial_\mu$ turns the momentum-space equation into the Dirac equation on spacetime,

$$\gamma^\mu \partial_\mu \Psi = -m\Psi.$$

A plane wave $\Psi(x) = e^{ip_\mu x^\mu} u$ solves it exactly when its amplitude u solves the momentum-space equation at p. The solutions of one mass form a complex vector space, so every finite superposition of on-shell plane waves of mass m is again a solution. The phase $p_\mu x^\mu$ is the de Broglie phase, and the amplitude is the on-shell spinor assembled from a dual spinor. For any two differentiable solutions of the same real mass the current $j^\mu = \bar{\Psi}\gamma^\mu \Phi$ is conserved,

$$\partial_\mu j^\mu = 0,$$

and between two plane waves of one momentum it is uniform. For $\Phi = \Psi$ the time component $j^0 = i\Psi^\dagger \Psi$ makes $\rho = \Psi^\dagger \Psi$ a density that is nonnegative, vanishes only where Ψ does, and obeys the continuity equation $\partial_t \rho + \nabla \cdot J = 0$ with the real spatial current $J^k = -ij^k$. This is the density that Born's rule requires. It is carried by the Dirac current, not by the Killing self-pairing, which is indefinite.

A wave of one momentum does not locate a chart point. The density of $e^{ip_\mu x^\mu} u$ is $\|u\|^2$ at every x, and the spinor at one point is a phase of modulus one times the spinor at any other. If the amplitude does not vanish, the wave vanishes nowhere. A field multiplied by one such phase under translation inherits that rigidity: one zero forces it to vanish at every chart point. A nonzero solution of one nonzero mass determines its four-momentum, and the same spinor does not solve the momentum-space equation at two distinct momenta. The sum of two plane waves of distinct momenta, with neither amplitude zero, fails to be a wave of any one momentum, and a translation does not multiply it by one position-independent phase. Where the two amplitudes are the same vector u, which nonzero mass excludes on the shell, the density is $4\cos^2\big((q-p)\cdot x/2\big)\,\|u\|^2$, and it vanishes where the relative phase is an odd multiple of π. Multiplication by $e^{ik_\mu x^\mu}$, which shifts the momentum by k, fails to commute with translation of the argument by a: the two orders differ by $e^{ik_\mu a^\mu}$. Whenever $k \neq 0$ and the amplitude does not vanish, the orders are not the same field. Dividing out the phase $e^{ip_\mu x^\mu}$ returns the amplitude at every chart point. The same division, applied to P_μ of the wave, returns p_μ times the amplitude: the momentum operator multiplies the carrier and does not turn it by a dual rotor. On every differentiable field the position operator, which multiplies by the chart coordinate x^ν, and the momentum operator satisfy $[X^\nu, P_\mu] = i\delta^\nu_\mu$. The coordinates commute with one another, and on a twice differentiable field so do the momenta, so this is the only nonzero bracket of the chart pair. Along $x + ta$ the derivative of a differentiable field at $t = 0$ is $ia^\mu P_\mu \Psi$, and on a momentum eigenfield it is $i(p \cdot a)$ times the field. A constant total rotor that acts on the value and leaves the argument fixed commutes with every position component and every momentum component: the chart pair does not turn the carrier. The orbital momentum $L_3 = X^1 P_2 - X^2 P_1$ built from that pair turns the transverse coordinates, $[L_3, X^1] = iX^2$ and $[L_3, X^2] = -iX^1$, while X^3 commutes with L_3, and so does the same rotor. A position eigenfield vanishes off one coordinate hyperplane, or at every chart point when the eigenvalue is not real. A momentum eigenfield with one zero vanishes everywhere, so the only joint eigenfield is zero. A differentiable field is an eigenfield of every P_μ, with eigenvalues p_μ, exactly when a translation multiplies it by one phase $e^{ip_\mu a^\mu}$. A nonzero field of that kind, and in particular a plane wave of nonzero amplitude, is an eigenfield of no X^ν, and its density does not depend on the chart point. A continuous eigenfield of any one X^ν vanishes at every chart point. The same trace that excludes the canonical pair from the dual sector excludes it from a single Dirac value. An uncertainty bound for orbital angles remains open.

The total-rotor covariance extends to spacetime. The field

$$\Psi'(x) = S(M)\,\Psi\big(\Lambda(M)^{\mathsf{T}} x\big)$$

solves $\gamma^\mu \partial_\mu \Psi' = -m\Psi'$ whenever Ψ does. The transposed action on the argument is fixed by the phase: $p_\mu x^\mu$ evaluated at $\Lambda(M)^{\mathsf{T}} x$ is $(\Lambda(M)p)_\mu x^\mu$, so the plane wave $e^{ip_\mu x^\mu} u$ goes to the plane wave of momentum $\Lambda(M)p$ with amplitude $S(M)u$. The de Broglie phase is transported by the usual factor; a dual rotor fixes the rest momentum and leaves the phase of a rest-frame plane wave in place.

4.7 Electromagnetic coupling

An electromagnetic potential A_μ enters through the covariant derivative $D_\mu = \partial_\mu - ieA_\mu$, and the coupled equation is

$$\gamma^\mu D_\mu \Psi = -m\Psi.$$

It is gauge covariant: for every differentiable χ, the pair $\Psi \mapsto e^{ie\chi}\Psi$, $A_\mu \mapsto A_\mu + \partial_\mu\chi$ carries solutions to solutions. The gauge factor is a scalar U(1) phase. It commutes with every total rotor and is neither a dual rotor nor its character. In a constant potential a the plane wave $e^{ip_\mu x^\mu}u$ solves the coupled equation exactly when u solves the momentum-space equation at the kinetic momentum $p - ea$: the canonical momentum p carries the phase, and the kinetic momentum carries the mass shell and the spinor assembled from a dual spinor. The current $\bar\Psi\gamma^\mu\Phi$ is unchanged by a gauge transformation of both spinors, and for real mass, charge, and potential it remains conserved on solutions of the coupled equation. The Born density therefore survives the coupling.

For twice-differentiable spinors the covariant derivatives fail to commute by the field strength,

$$[D_\mu, D_\nu]\Psi = -ieF_{\mu\nu}\Psi, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Applying $\gamma^\nu D_\nu$ to the coupled equation and splitting $\gamma^\mu\gamma^\nu$ into its symmetric and antisymmetric parts, every twice-differentiable solution obeys

$$\eta^{\mu\nu}D_\mu D_\nu\Psi - \frac{ie}{2}\,F_{\mu\nu}\gamma^\mu\gamma^\nu\Psi = m^2\Psi.$$

At zero charge this is the Klein–Gordon equation $\eta^{\mu\nu}\partial_\mu\partial_\nu\Psi = m^2\Psi$ for every component of a free solution. The field-strength term is block diagonal in the Weyl representation. With the magnetic field $\mathbf{B} = (F_{23}, F_{31}, F_{12})$ and the electric field $E_k = F_{k0}$, it acts as

$$e\,\sigma\cdot(\mathbf{B} + i\mathbf{E}) \text{ on the upper block}, \qquad e\,\sigma\cdot(\mathbf{B} - i\mathbf{E}) \text{ on the lower block}.$$

The magnetic part is the same on both blocks. Since $\sigma\cdot\mathbf{B} = i\sum_a B_a(-i\sigma_a)$ and $-i\sigma_a$ are the dual generators, the magnetic field acts on the spinor through the dual sector, in the way a dual rotor acts by one matrix on both blocks. The electric part changes sign between the blocks, as the generators of a usual rotor do. The magnetic field therefore couples to the spin carried by the little group, and the electric field to the boost carried by the usual sector. With $\Pi_k = -iD_k$ the spatial derivatives and the magnetic term combine into $-(\sigma\cdot\Pi)^2$, because $(\sigma\cdot\Pi)^2 = \Pi^2 - e\,\sigma\cdot\mathbf{B}$ follows from the commutator. The magnetic moment thus enters with gyromagnetic ratio 2, the value fixed by the first-order equation. The same split joins the wave to the internal spin. The orbital square stays with the chart, and the magnetic piece is the dual generator of Section 8.

4.8 The dual map, charge conjugation, and the antiparticle

The dual map $X \mapsto Xi$ multiplies by the pseudoscalar $i = e_0e_1e_2e_3$. On Dirac spinors that pseudoscalar is

$$\omega = \gamma^0\gamma^1\gamma^2\gamma^3 = i\gamma^5,$$

which is $+i$ on the upper Weyl block and $-i$ on the lower, so $\omega^2 = -1$. It anticommutes with every γ^μ and commutes with every total rotor $S(M)$. It therefore carries a solution of mass m at p to a solution of the same mass at $-p$. On spinors the reversal of time orientation by the dual map is thus the reversal of the four-momentum, and with it of the sign of the energy. At rest it sends the pair (u, iu) to $i(u, -iu)$, the rest solution at $(-m, 0, 0, 0)$. It leaves the current and the density $\Psi^\dagger\Psi$ unchanged and reverses the sign of the mass bilinear. On spacetime the field $\omega\Psi(-x)$ solves the coupled equation of the same charge in the potential $-A_\mu(-x)$, the image of A_μ under the point reflection $x \mapsto -x$, and it carries the plane wave of momentum p to the plane wave of momentum $-p$.

The dual spinor carries a second structure. The map

$$Ju = i\sigma_y\,\bar u$$

is antilinear with $J^2 = -1$, preserves the norm, and makes Ju orthogonal to u. It commutes with every dual rotor and reverses the spin along every axis, $J(\sigma\cdot n\,u) = -\sigma\cdot n\,Ju$, so it is the time reversal of a spin-$\frac{1}{2}$ state, and with it $\mathbb{C}^2$ is a quaternionic line on which the dual rotors act. For a total rotor it exchanges the actions on the two Weyl blocks, $J(Mu) = (M^\dagger)^{-1}Ju$. Charge conjugation

$$\Psi^c = (-Jl,\ Ju) \qquad \text{for } \Psi = (u, l)$$

is therefore an antilinear involution that commutes with every γ^μ and every total rotor. Because it is antilinear it reverses the sign of im, so it is a bijection between the solutions of mass m at p and at $-p$. In a potential it reverses the charge: Ψ^c solves the equation of charge $-e$ in the same potential A_μ, and

the plane wave $e^{ip_\mu x^\mu} u$ goes to the negative-frequency wave $e^{-ip_\mu x^\mu} u^c$. It preserves the density and the real current of a single spinor and reverses the sign of the mass bilinear.

Neither operation alone produces a positive-energy solution of opposite charge. Charge conjugation reverses the charge and the energy, and the dual map reverses the energy and keeps the charge. Their composite, the antiparticle

$$\Psi \mapsto \omega \Psi^c,$$

commutes with every total rotor and carries a solution of mass m at p to a solution of the same mass at the same p. For a single spinor it keeps the density, the current, and the mass bilinear. On spacetime the field $\omega \Psi^c(-x)$ solves the equation of charge $-e$ in the point-reflected potential $-A_\mu(-x)$, and it carries the plane wave of momentum p to the plane wave of the same momentum with amplitude ωu^c. At rest the antiparticle of (u, iu) is $-(Ju, iJu)$, the rest pair of the dual spinor Ju, with the spin reversed along every axis. The composite is antilinear and reverses all four coordinates, as the CPT operation of the Dirac equation does.

This is the reading of Stueckelberg and Feynman [17, 18]: the charge conjugate is a negative-frequency solution of opposite charge, a particle running backward in time, and reversing the time orientation turns it into a positive-energy solution of opposite charge running forward. In double spacetime both steps are already in the algebra. The charge is reversed by the quaternionic structure of the dual spinor, and the time orientation by the dual map. The antiparticle is the charge conjugate read through the dual spacetime.

5 Two Dual Spinors and Entanglement

The spin of a massive particle is carried by its dual spinor, the little-group parameter of Section 4. Two particles therefore carry two dual spinors, and a joint state is a vector of $\mathbb{C}^2 \otimes \mathbb{C}^2$. Written as the array Ψ_{ij} of amplitudes of $e_i \otimes e_j$, it is acted on by a rotor A on the first particle and a rotor B on the second as

$$\Psi \mapsto A \, \Psi \, B^{\mathsf{T}},$$

and the product states $u \otimes v$ are the arrays $u_i v_j$.

The antisymmetric invariant $\det \Psi$ separates the product states from the rest. A pair of rotors multiplies it by $\det A \det B$, so it is unchanged when each particle is turned by its own total rotor, usual or dual. It vanishes exactly on product states, and

$$2\,|\det \Psi| \le \|\Psi\|^2.$$

Dual rotors are unitary and also keep the norm, so the ratio $2|\det \Psi|/\|\Psi\|^2$, which is 0 on product states and at most 1, is unchanged by independent dual rotations of the two particles. No local dual rotor creates or removes entanglement.

The joint states left fixed by every common dual rotor $U \otimes U$ form a single line, spanned by the singlet

$$\varepsilon = e_0 \otimes e_1 - e_1 \otimes e_0.$$

A common total rotor $M \otimes M$ multiplies ε by $\det M = 1$, so the singlet is fixed by the usual factor as well. It is not a product, since $\det \varepsilon = 1$, and it attains the bound above. The singlet is built from the quaternionic structure J of Section 4.8: for every unit dual spinor u,

$$Ju \otimes u - u \otimes Ju = \varepsilon.$$

The right-hand side does not depend on u. The singlet pairs each dual spinor with its spin-reversed partner and singles out no axis. Its overlap with a product state $u \otimes v$ is $-\langle Ju, v\rangle$. That overlap vanishes when the two spins are parallel, and for the reversed partner $v = Ju$ it is $-\|u\|^2$.

On a unit joint state each particle has polarization $\sqrt{1 - C^2}$ with $C = 2|\det \Psi|$: a product leaves each factor with its own spin vector, and the singlet leaves neither particle a direction of its own. Exchange of the two particles, $\Psi \mapsto \Psi^{\mathsf{T}}$, commutes with every common rotor. It splits the joint space into the line of the singlet, on which the total spin squared vanishes, and the symmetric states, on which that square is 2. The orthogonal complement of the ray through a nonzero dual spinor u is the ray through Ju. The partner has the same norm and the opposite spin vector, and the two rays exhaust the dual spinor space, so no third ray is orthogonal to both. Two dual spinors on the same ray carry one spin vector. Their antisymmetrized product vanishes, so they occupy only the symmetric state of total spin 1. The singlet

is the joint state of a ray with its reverse. That is the exclusion of a repeated dual spinor. It is not the spin-statistics theorem, and spatial overlap of the repeated ray remains the repulsive torsional layer.

A spin measurement along a unit axis a is the pair of projectors $(1 \pm \sigma \cdot a)/2$. In the normalized singlet each particle alone has vanishing spin mean along every axis. The correlation of measurements along a on the first particle and along b on the second is

$$E(a, b) = \langle \sigma \cdot a \otimes \sigma \cdot b \rangle = -a \cdot b.$$

The outcomes $s, t = \pm 1$ occur with probability $(1 - st\, a \cdot b)/4$, so along axes at angle θ the two outcomes are opposite with probability $\cos^2(\theta/2)$. The angle enters at half its value, as it does in the dual rotor $U(\beta)$ of Section 3.

No local action on one particle is visible at the other. For every joint state, a dual rotor applied to the first particle leaves every expectation of the second unchanged. So does the choice of the axis measured on the first particle: summing over its two outcomes returns the statistics of the second particle alone, whatever the axis. The correlations of an entangled pair carry no signal.

They are nevertheless not correlations of particle-local values. Suppose each outcome is fixed in advance by variables of its own particle, such as its dual phases, and that these variables are distributed over the pair in any way. Then the combination

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b')$$

of Clauser, Horne, Shimony, and Holt obeys $|S| \leq 2$ [44, 45]. For the singlet, with a and a' perpendicular and b, b' their two bisectors, $S = -2\sqrt{2}$, and no choice of unit axes gives the singlet a larger $|S|$. The correlation that Einstein, Podolsky, and Rosen took as a sign of incompleteness [46] is in double spacetime a property of the joint dual state. It is not a pair of correlated dual phases, one carried by each particle. The singlet is the joint state that common dual rotation leaves unchanged, and in it neither particle has a spin direction of its own.

6 The Bound Electron

The electron of Section 4 is free: a dual spinor on the mass shell, the pair (u, iu) at rest. Binding it to a charge sets the two electromagnetic phases equal. The scalar interference factor of that common phase has exactly one zero in each interval of length π in the half-rapidity. Those zeros are force nodes, ordered by the rapidity. Distance enters only by inverting the common phase, so a phase that decreases strictly with distance places the next node inside the previous one, at a radius ratio less than 1.

The Bohr radii $r_n = n^2 a_0$ increase with n, and the first two stand in the ratio 4. No such phase places two successive nodes on two successive members of that sequence. In particular a_0 and $4a_0$ are not the radii of the first two nodes, and the hydrogen spectrum is a different sequence from the equal-scale nodes.

A circular speed $v^2 = ke^2/(m\gamma_s r)$ is real only where the interference factor is positive. A node is a divergence of that force, and the barrier does not quantize the orbit. The equal-scale geometry fixes the inward order of the barriers and the sign of each layer. The absolute scale of the barriers is the electromagnetic length. One revolution of the dual rotor restores every point of its orbit and leaves the rotor at -1, at every radius. The plane-wave factor on that revolution equals 1 exactly when $pr = n\hbar$. The circular path fixed by that return lies at $r = n^2 a_0 \gamma_s$, strictly inside $n^2 a_0$ by a fraction less than $1/60000$. The phase x of that path is the unique solution of $x\gamma_s(x) = \alpha/(2n^2)$ below $1/200$, on the restoring side, and the interference node is not that path. Read at $\gamma_s = 1$, every shell lies in the outer well, on the restoring side of a circular path. The step from shell 2 to shell 1 is $3/4$ of the ionisation, from shell 3 to shell 2 is $5/36$, from shell 4 to shell 2 is $3/16$, from shell 5 to shell 2 is $21/100$, and from shell 4 to shell 3 is $7/144$. These are the Lyman α line, the Balmer lines Hα, Hβ and Hγ, and the first Paschen line. Each of those differences departs from its fraction by less than one part in ten thousand of the fraction. At each such shell the outer-well potential lies strictly below its Coulomb value $-2/n^2$ of the ionisation, by at most $(3/5)\alpha^2/n^6$. On the whole exterior of the Bohr radius the same deepening is less than $1/30000$ of the ionisation, and the potential keeps falling through the outer well.

On the SI stand-ins, with the proton fixed, the five hydrogen wavelengths lie strictly between 121502 and 121503, between 656112 and 656113, between 486009 and 486010, between 433936 and 433937, and between 1874606 and 1874607, in units of 10^{-12} m. Shared inertia, with the proton-to-electron mass ratio strictly between 1836 and 1837, places them strictly between 121568 and 121569, between 656469 and 656470, between 486273 and 486274, between 434172 and 434174, and between 1875626 and 1875628, in the same units. This series is hydrogen, one proton and one electron, and the line of any other nucleus is not read by scaling these steps.

7 Torsional Stiffness and the Higgs Reading

Section 4 gives the momentum-space Dirac equation as a Clifford factor of $Q(p)$. What follows is a geometric hypothesis for the *value* of m, not a derivation of that value from torsion.

The torsional mismatch between the usual-sector rotor R_{usual} and the dual-sector rotor R_{dual} is quantified by the relative angle $\delta_a = \phi_a - \theta_a$. A modelling stiffness proportional to the rest mass m may be assigned to that angle. In the low-energy limit the associated potential is

$$V_{\text{torsion}} = \frac{1}{2} m \sum_{a=1}^{3} \delta_a^2.$$

This V_{torsion} is a particle-intrinsic oscillator potential, not the parameter-space Killing quadratic of the main paper, and it is not a restoring force in the Euler–Lagrange system of the written opposite-sign action of Section 8. The bilinear

$$m\bar{\psi}\psi \propto \text{Tr}(\Omega_{\text{biv}}),$$

with $\Omega_{\text{biv}} = \log(R_{\text{usual}}^{\dagger} R_{\text{dual}})$, is a geometric reading of the Dirac mass, not a derivation of m from the lag.

The harmonic stiffness is not a Higgs mechanism. In the Standard Model the Higgs field acquires a vacuum expectation value away from the symmetric point, and the curvature there generates masses. Here the written potential is minimized at alignment. Under the opposite kinetic signs of the particle action every constant lag is free, and the lag is inertial only where the slope vanishes, so a nonzero harmonic lag accelerates the common rapidity rather than settling into a misalignment. A quartic well off the origin can freeze a nonzero lag, but only when the two kinetic signs agree; the common phase is then flat, and its rate is initial data, not a mass for W or Z. No Weinberg angle is produced. On one axis alignment is $J = \frac{1}{2}\delta\sigma = 0$ with $M = \sigma^2/4$, the vacuum lies only at the origin, and a frozen well at rest is a balanced state of mass $v^2/4$.

8 Unified Dynamics of Quantum Mechanics and Gravity

The rotor Lagrangian is a model for the six rapidities, not a deduction from unitarity of the rotors. Measurement as synchronization, the torsional reading of entanglement, and the path integral below remain geometric hypotheses. The quantum side is the kinematics of the dual rotor, and usual–dual misalignment is the gravitational side.

Consider a single massive particle whose complete rotor is $R_{\text{total}} = R_{\text{usual}} R_{\text{dual}}$. The written effective action for the intrinsic rotor angles is

$$S_{\text{particle}} = \int \left[\frac{1}{2} \sum_{a=1}^{3} \dot{\phi}_a^2 - \frac{1}{2} \sum_{a=1}^{3} \dot{\theta}_a^2 - \frac{m}{2} \sum_{a=1}^{3} (\phi_a - \theta_a)^2 \right] dt.$$

The opposite signs of the kinetic terms are part of the written model. They are intended to reflect the boost-like (hyperbolic) character of the usual sector and the rotation-like (elliptic) character of the dual sector; they are not deduced from unitarity of the rotors. The last term is the oscillator potential of Section 7, not the parameter-space Killing quadratic of the main paper.

On one axis the density is equivalently the mixed form

$$L = \tfrac{1}{2}\dot{\sigma}\dot{\delta} - \tfrac{m}{2}\delta^2,$$

with common rapidity $\sigma = \phi + \theta$ and mismatch $\delta = \phi - \theta$. The Euler–Lagrange equations are

$$\ddot{\phi}_a = -m(\phi_a - \theta_a), \qquad \ddot{\theta}_a = -m(\phi_a - \theta_a).$$

Both angles accelerate together. The mismatch is therefore free, $\ddot{\delta}_a = 0$, and what the potential sources is the common rapidity, $\ddot{\sigma}_a = -2m\delta_a$. On one axis the general solution is the affine lag $\delta(t) = \delta_0 + \nu t$ together with the cubic common rapidity

$$\sigma(t) = \sigma_0 + \dot{\sigma}_0\, t - m\delta_0\, t^2 - \frac{m\nu}{3}\, t^3.$$

A frozen lag accelerates σ uniformly; a nonzero rate makes the acceleration itself linear in time. The mass, written as a stiffness of δ, is a force on σ and does not restore a frequency [4].

The Jacobi integral

$$E = \tfrac{1}{2}\dot{\phi}^2 - \tfrac{1}{2}\dot{\theta}^2 + \tfrac{m}{2}\delta^2 = \tfrac{1}{2}\dot{\sigma}\dot{\delta} + \tfrac{m}{2}\delta^2$$

is conserved along those equations. Along the motion above it keeps its initial value $\frac{1}{2}\nu\dot{\sigma}_0 + \frac{m}{2}\delta_0^2$: the quadratic growth of $\dot{\sigma}$ cancels the quadratic growth of the potential, so E stays fixed while the rapidity runs away. The integral is indefinite: a purely dual velocity at vanishing lag already yields $E < 0$. Conservation of an unbounded quadratic does not select a frequency.

A neighbouring system sometimes written alongside the action,

$$\ddot{\phi}_a = m(\phi_a - \theta_a), \qquad -\ddot{\theta}_a = m(\phi_a - \theta_a),$$

makes the mismatch run away, $\ddot{\delta}_a = 2m\delta_a$. It is not an oscillator.

An oscillator appears only if both kinetic signs are taken positive. The density $\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}\dot{\theta}^2 - \frac{m}{2}(\phi-\theta)^2$ yields $\ddot{\delta} + 2m\delta = 0$, at frequency $\sqrt{2m}$ in units $\hbar = c = 1$, while the common rapidity is free. The general solution is $\delta(t) = A\cos(\sqrt{2m}\,t) + B\sin(\sqrt{2m}\,t)$, with $\delta(t)^2 \le A^2 + B^2$, and its Jacobi integral equals $\frac{1}{4}\dot{\sigma}_0^2 + \frac{m}{2}(A^2 + B^2)$. For $m \ge 0$ that integral is nonnegative, so conservation does bound the motion of δ. The frequency $\sqrt{2m}$ and the Compton frequency m meet only at the isolated values $m = 0$ and $m = 2$, and the model is not the written action. The neighbouring sign pattern $\ddot{\delta} = 2m\delta$, at the same rate, is instead the unbounded hyperbolic motion $A\cosh(\sqrt{2m}\,t) + B\sinh(\sqrt{2m}\,t)$.

The written action therefore supplies no harmonic bridge to quantum theory. Identifying the rate ν of free lag with $E/\hbar$ remains a reading; the accumulated lag may then be written

$$\int \delta_a \, dt \propto \frac{Et - \mathbf{p}\cdot\mathbf{x}}{\hbar},$$

the de Broglie phase of the matter wave [3]. The same phase carries the plane-wave solutions of the spacetime Dirac equation of Section 4; what remains a reading is its identification with the free lag. The scalar U(1) factor built from δ is distinct from the real character of the dual rotor and from its spinor matrix element. The return of the plane-wave factor on one revolution is the condition $pr = n\hbar$ of Section 6.

The dual generator $\frac{1}{2}\,\omega\cdot\sigma$ evolves a dual spinor by the dual rotor $U(t\omega)$. The curve $u(t) = U(t\omega)u_0$ solves

$$i\,\dot{u} = \tfrac{1}{2}\,(\omega\cdot\sigma)\,u.$$

The parameter t scales the dual rapidity. Along this curve the time component the chart receives remains the information weight $\|u\|^2$. When the energy component p_0 is nonzero, the same real number, read as a chart time in the de Broglie phase, gives that phase the slope p_0. These are not the same motion. A chart phase multiplies the identity, and for $m \neq 0$ the matrix $-(e/2m)\,\sigma\cdot\mathbf{B}$ is traceless, so the magnetic spin Hamiltonian is never a chart phase.

What joins the internal spin to the wave is the field strength of that wave. The spatial kinetic momenta are the covariant derivatives of Section 4, $\Pi_a = -iD_a$, read on the carrier. On a twice differentiable carrier in a differentiable potential,

$$(\sigma\cdot\Pi)^2 = \Pi^2 - e\,\sigma\cdot\mathbf{B},$$

with $\mathbf{B} = (F_{23}, F_{31}, F_{12})$. The square Π^2 is the orbital kinetic term of the wave. The remainder $-e\,\sigma\cdot\mathbf{B}$ is a dual generator, and it is the same matrix on both Weyl blocks. The electric part of the same field strength changes sign between the blocks, and the two blocks receive one matrix exactly when the charge vanishes or the electric field does. For $m \neq 0$ the magnetic piece divided by $2m$ is the Hamiltonian $-(e/2m)\,\sigma\cdot\mathbf{B}$, which is $\frac{1}{2}\omega\cdot\sigma$ at the rapidity rate $\omega = -(e/m)\mathbf{B}$. The spin curve of that rate is the evolution this Hamiltonian generates, and along it the time component the chart receives remains the information weight. The angular speed is $|e|B/m$, the cyclotron frequency of a charge e and mass m in the field B, so the gyromagnetic ratio read off the split is 2.

A chart point is the argument of the phase $e^{ip_\mu x^\mu}$ of Section 4. The matrix obtained by multiplying the identity by a phase $e^{i\alpha}$ is a dual rotor exactly when that phase is ± 1. A constant external potential supplies the dual rotor of phase e^{-iVt} at every time exactly when $V = 0$, and a nonzero momentum kick meets a chart point whose phase is other than ± 1. What is placed first is the carrier, and an external position and an external time are this phase's argument. The domain of the Schrödinger equation lies on the side of that argument, and so do the position and momentum operators: one multiplies a field by the argument, and the other differentiates it.

The outer-well integral of the signed force lies on a radius placed after the carrier. On every radius at least the Bohr radius, the integral lies strictly below the Coulomb potential, and the deepening is less

than 1/30000 of the ionisation. At each shell $r_n = n^2 a_0$ the deepening of the Coulomb value $-2/n^2$, counted in the ionisation, is at most $(3/5)\alpha^2/n^6$. The integral keeps falling as the phase rises, and the fall is unbounded at the first node, so the shells lie on the descent. Read as an evolution on that radius, compactness of the dual sector does not supply the unitarity from the written particle action.

Wave-function collapse is readable as forced synchronization of dual rotors across an entangled many-particle system. A measurement interaction that coupled to the usual spacetime would still have to supply a restoring mechanism: the written action does not drive δ_a toward zero, and minimizing a global J is a further identification, not an Euler–Lagrange consequence. The apparent non-unitary jump would then be the classical limit of a rapid, collective torsional relaxation. No additional collapse postulate is required in that picture; the picture itself remains a hypothesis.

The kinematics of entanglement is fixed by Section 5, and a torsional reading has to respect it. The correlation of an entangled pair cannot be a correlation of dual phases carried separately by the two particles, since those keep the CHSH combination within 2 and the singlet reaches $2\sqrt{2}$. It lives in the joint dual state. A local dual rotation of one particle changes neither that entanglement nor any statistics of the partner, so the correlation transmits nothing, and information still propagates at most at the speed of light through the usual sector. What remains a reading is the energy of the pair: a potential for its collective mismatch, which the written action does not supply, and its relaxation as the mechanism of collapse. Compactness of the dual sector makes that collective mismatch available as a geometric option, not as a theorem of the written action.

A path-integral formulation may be written with the same particle action,

$$\langle R_f | R_i \rangle = \int \mathcal{D}[\phi, \theta] \, \exp\left(\frac{i}{\hbar} S_{\text{particle}}[\phi, \theta] \right),$$

the integral running over paths in the dual-rotor phase space. Compactness of the dual sector would keep the integral well-defined. Because the written S_{particle} makes δ free, the torsional potential term does not by itself generate Compton-frequency interference.

In this picture quantum mechanics and gravity are not separate theories that must be reconciled. Quantum mechanics is the kinematics of rotations in the compact dual spacetime; gravity is the dynamical cost of breaking the parallelism between the usual and dual rotors. Both readings originate from the same 16-dimensional biquaternion algebra. The kinematics is sharp. The dynamical bridge from the written particle action is not.

9 Dual Spacetime as Intrinsic Supersymmetry

The cosmological constant problem, widely known as the vacuum catastrophe, represents one of the most acute fine-tuning crises in theoretical physics. Quantum field theory predicts that the vacuum energy density arises from the zero-point fluctuations of every field mode up to the ultraviolet cutoff. With the cutoff taken at the Planck scale, this yields an energy density of order $M_{\text{Pl}}^4 \approx 10^{74}\,\text{GeV}^4$, which converts in SI units to approximately $10^{113}\,\text{J/m}^3$. The observed value, however, inferred from the present-day accelerated expansion, is only $\rho_{\text{vac}}^{\text{obs}} \approx 5.4 \times 10^{-10}\,\text{J/m}^3$, corresponding to a cosmological constant $\Lambda \sim 1.1 \times 10^{-52}\,\text{m}^{-2}$. The ratio between the theoretical prediction and the measured value therefore reaches

$$\frac{\rho_{\text{vac}}^{\text{theory}}}{\rho_{\text{vac}}^{\text{obs}}} \sim 10^{120}.$$

No known renormalization procedure can remove this discrepancy without an extraordinary cancellation between bare parameters and quantum corrections, accurate to one part in 10^{120}. The problem is not technical but conceptual: gravity cannot be consistently coupled to the vacuum of quantum field theory on a fixed background without destroying the observed cosmology.

The Double Spacetime Theory addresses the crisis from inside the algebra: supersymmetry is already realized *intrinsically and inseparably* in the 16-real-dimensional biquaternion algebra carried by every massive particle. No new fields or particles are introduced, and no external symmetry is imposed or broken.

The six generators of the underlying Lorentz candidate $\mathfrak{so}(3,1)$ (not the twelve-dimensional sum $\mathfrak{so}(3,1) \oplus \mathfrak{so}(3,1)$) partition into two sectors distinguished by the sign of their quadratic form: the hyperbolic sector, whose generators square to $+1$, and the cyclic sector, whose generators square to -1. Same-axis generators commute; off-axis commutators need not vanish. The complete rotor R_{total} acts on both sectors simultaneously. The dual map $X \mapsto Xi$ interchanges the two sectors up to signs while

reversing time orientation. This interchange is an internal automorphism of the algebra; it does not map a particle onto a distinct species but exchanges two permanently coupled aspects of one and the same 16-dimensional object. The supersymmetric partner of any bosonic excitation is therefore already present as the dual-sector component, and there is no external superpartner to discover. On Dirac spinors the dual map acts by $\omega = i\gamma^5$ and reverses the four-momentum. Composed with the charge conjugation that the quaternionic structure of the dual spinor supplies, it carries every solution to its antiparticle at the same momentum (Section 4.8). The antiparticle is thus internal as well: it is the same Dirac field read through the dual map, not an additional field.

The scalar J is the parameter-space quadratic $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$ built from the mismatch bivector Ω_{biv}. Because the Killing form takes opposite values on the hyperbolic sector $(+8)$ and the cyclic sector (-8), bosonic and fermionic contributions enter with opposite signs, and they cancel when the two sectors are aligned. In this reading the residual vacuum value of J comes from fluctuations of the dual-sector angles. The cyclic generators square to -1, rendering the dual spacetime elliptic: the dual rotors form the compact group SU(2) (Section 3). The angles are therefore periodic and discretely sampled, $\theta_a \in (2\pi/N)\mathbb{Z}$ for large integer N. Compactness does *not* force an algebraic upper bound of order N^{-2}. On admissible configurations $|J| \leq 3\pi^2/8$, or $|J_{\text{norm}}| \leq 1$ with $J_{\text{norm}} = \frac{8}{3\pi^2}J$. The discrete quantity $\varepsilon_N = 16/(3N^2)$ is a *lower* bound on the nonzero normalized height, not an upper bound: every non-rest admissible lattice point satisfies $|J_{\text{norm}}| \geq \varepsilon_N$. Vacuum suppression, if it occurs, must come from a dynamical cancellation between sectors, not from writing $|J| \leq J_{\max} \sim N^{-2}$.

The dual sector carries spin-1/2 kinematics: it is the little group of a massive particle (Section 4). A full identification of Standard Model fermions with dual phases is not claimed. The dual map $X \mapsto Xi$ is not spacetime supersymmetry of a ten-dimensional string: $\text{Cl}(3,1) \ncong \text{Cl}(9,1)$, the dual spinor space is not the Majorana–Weyl module of $\text{Cl}(9,1)$, and the Super-Poincaré generator count is not the ten generators of the projective geometric algebra. The cancellation of opposite signatures at alignment does not generate the observed fermion masses, and the admissible bound on J is not the electroweak scale.

In this framework supersymmetry is not an optional extension that may or may not be realized in nature. It is the geometric consequence of an algebra whose two sectors possess quadratic forms of opposite signature and whose dual sector is compact. The vacuum catastrophe is therefore not met by adding new symmetries. The cancellation that observation requires has to come from the opposite signatures and the compactness already present in the 16-dimensional structure; the written particle action of Section 8 does not by itself produce it.

10 Concrete Predictions of Quantum Gravity

The double spacetime theory yields a series of sharp, falsifiable predictions that distinguish it decisively from both general relativity and other quantum gravity approaches. These predictions follow directly from the compactness of the dual spacetime and the resulting discretization of rotor angles.

10.1 Discrete Spacetime and Singularity Resolution

The dual rotor angles are discretized by the compact topology of the dual spacetime,

$$\phi_a, \theta_a \in \frac{2\pi}{N}\mathbb{Z},$$

with N an integer. The admissible ceiling

$$|J| \leq \frac{3\pi^2}{8}, \qquad |J_{\text{norm}}| \leq 1$$

is the bound of the admissible cone. Bivectors from many sources add without a ceiling; the configuration that enters J is the admissible point closest to that sum, so the ceiling still bounds the expressed mismatch of a composite source. Read as the particle's rest mass in Planck units, N places the nonzero normalized floor $\varepsilon_N = 16/(3N^2)$ at sixteen thirds of the gravitational fine-structure constant $Gm^2/\hbar c$. That reading produces a gravitational scale, the one compared with the electromagnetic constant for the electron in the companion paper. A separate estimate of the form $c^4/G\,\ell_P^{-2}$ remains a dimensionful curvature scale. The admissible ceiling forbids the unbounded curvature that appears at $r = 0$ in classical general relativity or at the Big Bang, provided the configuration remains in the admissible cone. Black hole and cosmological singularities are replaced by finite, multi-layered torsional configurations in which repulsive layers ($J < 0$) halt collapse before a true singularity can form.

On the admissible cone the normalized mismatch J_{norm} falls into four regimes. Balance is $J_{\text{norm}} = 0$. Unsaturated attraction is the open interval $0 < J_{\text{norm}} < 1$. Saturated attraction is the single point $J_{\text{norm}} = 1$: every axis is a pure boost, and it is the only admissible point whose usual rapidities admit no other dual completion. Repulsion is $-1 \leq J_{\text{norm}} < 0$, and the pure rotation at -1 remains inside that regime [1]. Dual-only motion holds the usual rapidities fixed and is frozen at saturated attraction, so the ceiling has no repulsive continuation until the usual sector moves. A repulsive continuation exists precisely when the dual wall is strictly negative, $\sum_a \alpha_a < 3\pi/4$, and a balanced continuation exists precisely when the wall is nonpositive. A balanced continuation of a nonzero usual seed has positive unsigned mass, and is not the vacuum. The sector swap sends saturated attraction to the pure rotation. The lesser of an admissible height and the height of its swap is balance when the mismatch vanishes, and repulsion otherwise.

The unsigned mass $M = \frac{1}{2}\sum_a(\alpha_a^2 + \beta_a^2)$ is the Euclidean size of the same rapidities, normalised by the same factor, $M_{\text{norm}} = \frac{8}{3\pi^2}M$. It is not the value of the Dirac mass. On an admissible configuration $M_{\text{norm}} \leq 1$. Equality holds exactly when every axis is a corner, a pure boost or a pure rotation. If n of the three axes are pure boosts, the normalised mismatch is the discrete height $(2n-3)/3$. A configuration that meets the mass ceiling therefore has J_{norm} equal to one of 1, $\frac{1}{3}$, $-\frac{1}{3}$, and -1. The height 1 is saturated attraction: one point, and it carries the whole mass ceiling. The height $\frac{1}{3}$ is unsaturated attraction, with exactly one axis a pure rotation, hence three points. The height $-\frac{1}{3}$ is repulsion of the opposite census, again three points. The height -1 is the pure rotation and remains inside repulsion. These eight corners are the whole mass ceiling. Balance does not meet it. A balanced configuration has $M_{\text{norm}} \leq 5/6$, and that bound is attained. Sector swap preserves the unsigned mass and exchanges n pure boosts with $3-n$, so the two heavy attractive heights trade places with the two heavy repulsive ones. The reading that a repulsive layer halts collapse remains a conjecture. What the cone fixes is the census: every configuration that saturates the unsigned mass sits at one of these four heights, and gravitational neutrality is strictly lighter.

10.2 Absence of Event Horizons

In the continuum limit, light rays follow null geodesics that terminate at an event horizon. In the discrete dual spacetime, however, light rays encounter only a finite number of torsional layers. Resonance tunneling through these layers remains possible at all depths. The classical Kerr outer horizon is reinterpreted as a torsional transition shell rather than a one-way membrane. Observers falling inward experience a sequence of attractive and repulsive torsional regions but never encounter a true horizon. This resolves the black hole information paradox and the firewall problem without invoking exotic Planck-scale physics.

10.3 Strong-Field Deviations

Gravitational time dilation and redshift receive discrete corrections. The proper time interval measured by a stationary observer at radial coordinate r is

$$\frac{\Delta\tau}{\tau} = \sqrt{1 - \frac{2GM}{c^2 r}}\left[1 + \mathcal{O}\left(\frac{\ell_P^2}{r^2}\right)\right].$$

For the Earth's core the correction is of order 10^{-20}, for neutron star surfaces approximately 10^{-10}, and near Sgr A* of order 10^{-5}. These deviations are in principle detectable with next-generation atomic clocks and very-long-baseline interferometry.

10.4 Gravitational-Wave Echoes

After a binary merger, the remnant torsion star possesses a stratified interior with alternating attractive and repulsive layers. Gravitational waves partially reflect at each torsional interface, producing a train of echoes. The time delay between successive echoes is

$$\Delta t \sim \frac{GM}{c^3} \approx 10^{-5}\left(\frac{M}{10M_\odot}\right)\ \text{s}.$$

The echoes carry a characteristic phase modulation determined by the dual-rotor spectrum. Detection of such echoes by LISA or the Einstein Telescope would constitute direct evidence for the discrete torsional structure of strong gravity.

10.5 Cosmological Implications

The early universe is described as a single primordial torsion star. The Planck-era torsional mismatch drives a brief de Sitter-like expansion without requiring an inflaton field. Reheating occurs through layer transitions that convert torsional energy into radiation. The resulting primordial power spectrum is naturally scale-invariant, arising from the discrete rotor modes rather than from quantum fluctuations of a scalar field. Dark matter is unnecessary: galactic rotation curves are explained by residual baryonic torsional misalignment at galactic scales. Dark energy emerges as the minimum non-zero vacuum torsional rigidity $J_{\text{vac}} > 0$ induced by quantum rotor fluctuations, yielding a cosmological constant of order $\Lambda \sim \ell_P^{-2}$.

All of the above predictions are falsifiable within the next decade. Non-observation of strong-field deviations at the predicted level, absence of gravitational-wave echoes, or detection of a true event horizon would rule out the framework. Conversely, positive detection of any of these signatures would confirm that spacetime is discrete, particle-local, and doubly temporal at the most fundamental level.

11 Comparison with Other Quantum Gravity Approaches

The double spacetime theory occupies a distinctive position among quantum gravity candidates. Most frameworks quantize or discretize an a priori continuum geometry. The results collected here are algebraic identities in $\mathrm{Cl}(3,1)$ and $G(3,1,1)$, not a derivation of quantum mechanics and gravity from that algebra. Below we compare the programme with existing approaches and record why it does not identify with a ten-dimensional string.

11.1 Loop Quantum Gravity

Loop quantum gravity (LQG) discretizes spacetime via spin networks and quantizes area and volume operators while retaining a background manifold and Riemannian curvature as fundamental. Singularities are resolved by a minimal eigenvalue of the area operator, yet the continuum limit remains challenging and the dynamics are encoded in complicated Hamiltonian constraints.

In contrast, the double spacetime theory imposes discreteness directly on the dual-rotor angles $\phi_a, \theta_a \in (2\pi/N)\mathbb{Z}$, without introducing a background manifold. Curvature is absent; gravity is the torsional mismatch of the usual and dual rotors, a bivector in the six-dimensional Lie algebra $\mathfrak{so}(3,1)$. Macroscopic geometry is that same mismatch, carried by each particle's own usual and dual rotors.

11.2 String Theory

String theory unifies forces by embedding gravity in a ten- or eleven-dimensional spacetime populated by vibrating strings and requiring supersymmetry. While mathematically elegant, it suffers from a vast landscape of vacua and lacks unique low-energy predictions.

The algebras $\mathrm{Cl}(3,1)$ and $\mathrm{Cl}(9,1)$ are not isomorphic, and the dual spinor space $\mathbb{C}^2$ is not the sixteen-real-dimensional Majorana–Weyl module of ten-dimensional spacetime. Unification inside four-dimensional $\mathrm{Cl}(3,1)$ is a statement about one algebra, not a proof that extra dimensions are physically absent. The dual map is an internal automorphism; it is not spacetime supersymmetry. Gravity, electromagnetism, and the nuclear forces as modes of a single rotor algebra remain hypotheses; within the algebra, the electromagnetic coupling of Section 4 already separates a dual-sector magnetic term from a usual-sector electric term.

No Calabi–Yau compactification is used; compactness of the dual sector at the particle level is a geometric hypothesis, not a string compactification.

11.3 Canonical and Ashtekar Approaches

The Ashtekar reformulation casts general relativity in terms of self-dual connections, enabling a Yang–Mills-like quantization. However, it inherits the continuum singularities of classical gravity and must confront the Wheeler–DeWitt equation with its notorious factor-ordering and regularization problems.

In the present framework no connection is introduced. Parallel transport is replaced by rotor exponentiation, and the Killing form on the torsional bivector replaces the curvature scalar. There is therefore no Wheeler–DeWitt constraint to solve; whether the finite rotor image supplies a complete quantization is part of the dynamical question left open in Section 8.

11.4 Emergent Gravity and Causal Sets

Emergent gravity approaches treat spacetime as a thermodynamic or entropic phenomenon, while causal set theory discretizes Lorentzian order. Both frameworks struggle to recover the exact Einstein equations and face fermion doubling or chirality issues.

The double spacetime theory reaches the Einstein equations through the teleparallel equivalence once a rotor-to-tetrad dictionary is fixed [7]. A commuting projector in $G(3,1,1)$ generates an eight-dimensional left ideal that splits by the null generator and is not read as an irreducible fermion. Discreteness is algebraic (Diophantine constraints on rotor angles) rather than set-theoretic, and causality is preserved by the lightlike $J = 0$ resonance condition. No thermodynamic analogy is invoked. Gravity on every scale is the torsional mismatch of each particle's own usual and dual rotors.

In summary, existing quantum gravity programs extend or quantize the continuum manifold while retaining curvature as fundamental. The present paper records algebraic identities in $\mathrm{Cl}(3,1)$ and $G(3,1,1)$ and discards several identifications that do not hold. It does not derive spacetime, all interactions, or a controllable theory of gravity from those identities.

12 Experimental Testability from the Quantum Perspective

Although the double spacetime theory is fundamentally algebraic, its quantum sector makes concrete experimental predictions that can be tested with current or near-future technology. The mismatch scalar $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$ depends only on the relative rotor. Dual-only motion cannot raise J; it conserves $J + M$, so every dual-only neutral configuration from a pure usual seed doubles unsigned mass. Reaching neutrality is a global budget, $\sum \alpha \le 3\pi/4$, not an isotropic per-axis condition: a single boosted axis can still be made neutral by parking dual rapidity on a silent axis. Repulsion requires a strictly negative wall, so a uniform usual rapidity $\pi/3$ keeps every dual-only continuation in unsaturated attraction, while $\pi/6$ reaches balance and repulsion and still never the attractive ceiling. The vacuum $J = M = 0$ is mixed-reachable and dual-only unreachable except from the origin. The dual-only increment of the dual angles is helicity-even in J: a circular Faraday wave does not displace the mean, while a circular potential drops it ponderomotively and leaves the cone from a vanishing dual seed. Whether laboratory fields supply a helicity-odd dual-channel force remains a conjecture.

12.1 Theoretical Coupling Mechanism

A circularly polarised electromagnetic wave is null at every phase, with vanishing four-phase mean of every linear Faraday coefficient. Superposing that wave onto the mismatch does not shift mean J. Among the six classical Faraday quadratics, the unique helicity-odd combination on that wave is the Poynting component $(E \times B)_z = \sigma E_0^2$. The same dual-only increment, read as $\theta \mapsto \theta + eA$, drops mean J by the helicity-even ponderomotive $\frac{1}{2}e^2\langle A^2\rangle$ and is not Maxwell-gauge invariant. Helicity drive of the particle mismatch is neither that increment nor a local scalar of F. The gauge-invariant electromagnetic coupling that the algebra does supply is the minimal coupling of Section 4, in which the magnetic field acts on the dual spinor through the dual generators. The working model takes the Poynting bilinear as the candidate source of a dual-phase drive, with right-handed polarisation ($\sigma = +1$) reinforcing attractive mismatch ($J > 0$) and left-handed polarisation ($\sigma = -1$) opposing it. The effective interaction is written

$$H_{\mathrm{int}} = \kappa\sigma E_0^2 \sum_a \theta_a \Gamma_a,$$

where E_0 is the electric-field amplitude and κ is a material-dependent coupling, neither of which is derived from the dual-rotor algebra. The phase-lock

$$\dot{\theta} = \kappa\sigma E_0^2 \sin(\omega t - \theta + \delta_\sigma)$$

is extra structure. The written particle action, given a dual-channel acceleration u, makes the mismatch a double integrator $\ddot{\delta} = -u$ and supplies no restoring frequency. Dual-only kinematics is a constrained slice of that action; without the constraint, both sectors move.

12.2 Proposed Near-Term Experiments

The most accessible platform exploits collective coherence in superconductors or Bose–Einstein condensates, where macroscopic numbers of particles ($N \sim 10^{26}$) can lock their dual rotors into a common phase. A concrete proposal is as follows:

- Suspend a microgram-scale superconducting test mass inside an asymmetric THz resonator (quantum cascade laser or free-electron laser cavity) with precise circular-polarization control.
- Monitor the apparent weight or inertial mass via a high-precision torsion balance or superconducting gravimeter while sweeping the frequency and helicity.
- Look for helicity-dependent changes in effective mass. A resonant peak at a proposed dual-rotor scale $\theta \sim c/(2\pi r_{\mathrm{comp}})$ would test the extra phase-lock structure, not the dual-only geometry of J, and not the helicity-even ponderomotive of the dual-only increment.

Predicted effect sizes, if the coupling exists, are $\Delta m/m \sim 10^{-2}$–10^{-5} at achievable intensities of 10^{12}–10^{15} W/m^2. The signature is distinctive: the effect must reverse sign when the polarization helicity is flipped and must vanish for linearly polarized or unpolarized radiation.

12.3 Falsifiability and Caveats

A null result at the 10^{-6} level after a systematic frequency and helicity scan would constrain the dual-rotor coupling strength, not the admissible-cone kinematics of J. Positive detection—particularly a helicity-reversible mass shift—would constitute strong evidence that gravity and inertia are torsional phenomena originating in dual spacetime. A resonance peak would further constrain the working oscillator or phase-lock, which the written action does not force.

The theory makes no claim of immediate technological breakthrough. The proposals above are modest, table-top experiments whose primary purpose is to test the hypothesis that dual-spacetime degrees of freedom couple measurably to laboratory electromagnetic fields. Success at this level would open the door to larger-scale gravitational engineering; failure would simply bound the parameters of the coupling model.

12.4 Broader Implications

If dual-rotor control proves feasible, the same mechanism offers pathways to nuclear transmutation and radioactivity neutralization by driving torsional layer transitions inside atomic nuclei. These applications, however, lie beyond the scope of the present quantum-gravity test program and will be addressed in future work.

In summary, dual-only kinematics already determine when J can be cancelled or reversed and at what unsigned-mass cost, and they allow the dual budget to be routed across silent axes. Whether nature supplies a laboratory dual force—and whether gravity is therefore a controllable torsional degree of freedom—is the experimental question of the coming decade.

13 The Algebraic Picture

The quantum sector of double spacetime theory is the kinematics of a compact dual spacetime already present in the 16-real-dimensional algebra $\mathbb{H} \oplus \mathbb{H} \cong \mathrm{Cl}(3,1)$. Gravity is the dynamical cost of breaking the parallelism between the usual and dual rotors that every massive particle carries. No background manifold is quantized, and the Hilbert-layer pairing is treated in [1].

The dual sector is elliptic. Its generators square to -1, satisfy $IJ = K$, and act on $\mathbb{C}^2$ by $-i\sigma_a$, and the dual rotors are exactly the group SU(2). A vector of this sector is the body of the information. The four-vector with time component $\|u\|^2$ and spatial part equal to the spin vector is future lightlike, every future lightlike four-vector arises so, a unit phase leaves it fixed, and a dual rotor turns only the spatial part. The usual chart therefore receives the state on the whole future light cone and nowhere else. The probability current of a Dirac spinor is the signed sum of two such images: one chirality stays null, and the rest pair, whose two images coincide, has the timelike current $(2\|u\|^2, 0, 0, 0)$. A superposition is not the sum of the two four-vectors: that sum may be timelike, the discrepancy spacelike, and a relative phase, invisible in either vector alone, turns the null direction. A unit dual spinor is a point of the unit sphere, a dual rotor turns its spin vector by the full angle, the 4π period is the intensity of $u + U(\beta)u$, and the Robertson–Schrödinger relation for two spin axes is an equality. Usual and dual generators commute on a common axis and need not commute off-axis. With the usual factor written first, every total rotor factors uniquely as $R_{\mathrm{total}} = R_{\mathrm{usual}} R_{\mathrm{dual}}$, and the dual factors are exactly the total rotors that fix the rest momentum. The little group of a massive particle, which carries its spin, is therefore the dual

sector, and the momentum is carried by the usual sector; composing perpendicular usual rotors leaves the Thomas–Wigner rotation in the dual sector.

The Dirac matrices are the grade-1 generators of the same algebra, distinct from the hyperbolic bivectors that build square-+1 projectors. The momentum-space Dirac equation is the first-order factor of the Minkowski quadratic, its solutions on the mass shell are assembled from a dual spinor, and the equation is covariant under every total rotor, in momentum space and on spacetime. The mass bilinear is the invariant pairing; the Euclidean overlap is p_0/m times it and, up to the factor i, the time component of the conserved current. That component, $\Psi^\dagger\Psi$, is the nonnegative density that Born's rule requires, which the indefinite Killing self-pairing is not. A nonzero solution of nonzero mass determines its four-momentum, and the plane wave of that momentum has one density at every chart point. The sum of two plane waves of distinct momenta fails to be a wave of one momentum, and translation of the argument differs from a shift of the momentum by the phase $e^{ik_\mu a^\mu}$. On differentiable fields, multiplication by x^ν and $-i\partial_\mu$ satisfy $[X^\nu, P_\mu] = i\delta^\nu_\mu$. The dual sector, and a single Dirac value, carry no such pair. The discrete dual circle carries a unitary transform of its amplitudes onto winding labels. The sums of squared moduli agree up to the factor N, and up to N^3 on the three dual angles. Shift and winding character differ by the phase $e^{-2\pi irs/N}$, convolution becomes a product of transforms, and the supports of a nonzero amplitude and of its transform have product at least N, at least N^3 on the three dual angles, with both bounds attained. In the electromagnetic coupling the magnetic field enters through the dual generators with gyromagnetic ratio 2 and the electric field through the usual generators, so spin precession is a dual rotation. The dual map acts on spinors by $\omega = i\gamma^5$ and reverses the four-momentum, which is its reversal of time orientation. The quaternionic structure of the dual spinor defines a charge conjugation that commutes with every total rotor and reverses the charge, and the composite of the two is the antiparticle at the same momentum, the charge conjugate read backward in time. Two dual spinors form a joint state in the tensor product, whose entanglement no local dual rotor changes. The singlet $Ju \otimes u - u \otimes Ju$ is the unique joint state fixed by common dual rotation; exchange splits the joint space into that singlet, of total spin 0, and the symmetric states, of total spin 1, and two dual spinors on one ray cannot occupy the singlet. The orthogonal complement of that ray is the spin-reversed partner, and the two rays exhaust the space. The correlation $-a \cdot b$ carries no signal and reaches $|S| = 2\sqrt{2}$, which no assignment of particle-local dual phases attains. None of this derives the value of m from torsion, and equal-scale interference nodes, which run inward, are not the Bohr radii.

Gravity is the mismatch of the two rotors, measured by $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$ with the admissible ceiling $|J| \le 3\pi^2/8$; the discrete quantity $\varepsilon_N = 16/(3N^2)$ is a lower bound on nonzero normalized height, not a ceiling. On the cone J_{norm} takes four regimes, from the pure rotation through repulsion and balance to unsaturated and saturated attraction, and dual-only motion cannot leave saturated attraction for repulsion. The lesser of a height and its sector swap is repulsive whenever the height is nonzero. The unsigned mass shares the ceiling. The configurations that meet it are eight corners, at the heights 1, $\frac{1}{3}$, $-\frac{1}{3}$, and -1; a balanced mass reaches at most five sixths of that ceiling. Vacuum suppression, if it occurs, must come from a dynamical cancellation between the hyperbolic and cyclic sectors. Identification of J with the teleparallel scalar is taken from the literature after a rotor-to-tetrad dictionary is fixed [7], and the dual map $X \mapsto Xi$ is an internal automorphism, not ten-dimensional spacetime supersymmetry.

The written opposite-sign particle action does not close the dynamical bridge from this kinematics to quantum theory. It frees the mismatch, $\ddot{\delta} = 0$, and sources the common rapidity, $\ddot{\sigma} = -2m\delta$; its Jacobi integral is indefinite, and an oscillator appears only when both kinetic signs are positive, at a frequency $\sqrt{2m}$ that meets the Compton frequency only at isolated masses. Identifying the free lag with the de Broglie phase, measurement as synchronization, the torsional reading of entanglement, and the path integral remain geometric hypotheses placed on this kinematics. Along the curve of a dual rotor the time component of the chart remains the information weight, and the domain of the Schrödinger equation is the phase argument that comes after the carrier. A constant external potential is a dual rotor at every time exactly when the potential vanishes. The magnetic spin Hamiltonian is never that phase. On a twice differentiable carrier $(\sigma \cdot \Pi)^2 = \Pi^2 - e\,\sigma \cdot \mathbf{B}$, and for $m \neq 0$ the magnetic piece divided by $2m$ is the dual rotor of rate $-(e/m)\mathbf{B}$, of angular speed $|e|B/m$. The continuum remains a classical approximation, and the 16-dimensional algebra is the setting of the quantum sector, not a finished theory of quantum gravity.

14 Conclusions

The double spacetime theory supplies an algebraic setting for quantum gravity, not a completed derivation. Quantum kinematics is the compact dual sector of $\mathrm{Cl}(3,1)$; the information lives there, and the

usual chart receives exactly the future light cone. A superposition is not the sum of those four-vectors. Gravity is the mismatch of usual and dual rotors. The algebra determines the dual rotors as SU(2); the spin vector of a unit dual spinor as a point of the unit sphere, turned by the full angle, with the 4π interference and the Robertson–Schrödinger equality; the Dirac equation as the Clifford factor of the Minkowski quadratic, with on-shell solutions built from a dual spinor; its covariance under every total rotor, with the dual sector as the little group and the Thomas–Wigner rotation as a dual factor; the spacetime equation with its conserved current and Born density, a wave of one momentum having one density at every chart point, the carrier returned by dividing out that phase, and the sum of two distinct momenta failing to be such a wave; multiplication by the chart coordinate and $-i\partial_\mu$ satisfying $[X^\nu, P_\mu] = i\delta^\nu_\mu$ on differentiable fields, that bracket the only nonzero one of the chart pair, with no such pair on the dual sector, a nonzero momentum eigenfield being an eigenfield of no position component, and a continuous eigenfield of one position component vanishing at every chart point; the time component of the chart remaining the information weight along the curve of a dual rotor; a phase times the identity as a dual rotor exactly at ± 1, so that a constant external potential is a dual rotor at every time exactly when the potential vanishes; on a twice differentiable carrier the square $(\sigma \cdot \Pi)^2 = \Pi^2 - e\,\sigma \cdot \mathbf{B}$, whose magnetic piece over $2m$ is the dual rotor of rate $-(e/m)\mathbf{B}$ and is never a chart phase; an electromagnetic coupling in which the magnetic field acts through the dual generators and spin precession is a dual rotation; the antiparticle as the charge conjugate read through the dual map, which on spinors reverses the four-momentum; and the singlet of two dual spinors, whose exchange split is spin 0 against spin 1, whose repeated ray cannot occupy the singlet, the orthogonal spin-reversed partner exhausting the space with it, and whose correlations carry no signal and exceed the CHSH bound of particle-local variables. It also fixes the admissible ceiling on J, the four regimes of the normalized mismatch on that cone, the dual-only rigidity of saturated attraction, the eight corners of the unsigned-mass ceiling, the five-sixths bound on a balanced mass, and the inward order of equal-scale interference nodes. It does not determine a Compton oscillator from the written action, the value of the mass. The outward radii are the return of the plane-wave factor on one revolution of the dual rotor, the revolution that restores the point and leaves the rotor at -1. Read at $\gamma_s = 1$, the Lyman α line, the Balmer lines Hα, Hβ and Hγ, and the first Paschen line lie in the outer well, on the restoring side of a circular path, and the interference there shifts each of them by less than one part in ten thousand. At those shells the outer-well potential lies strictly below the Coulomb value $-2/n^2$ of the ionisation, by at most $(3/5)\alpha^2/n^6$, and outside the Bohr radius the deepening stays below 1/30000 of the ionisation.

Predictions and laboratory proposals in the preceding sections remain conjectural, bounded by the admissible ceiling. The continuum remains a classical approximation, and the 16-dimensional algebra is the setting, not a finished theory of quantum gravity.

References

[1] Anonymous, "Dual Spacetime 4-Valued Logic (D4L): Amplitudes, Four States, Geometric Interference, and Dual Hilbert Space" (2026). `https://github.com/hypernumbernet/DstDiophantine/blob/main/papers/dst-4-valued-logic.tex`.

[2] Anonymous, "Gravity as Torsion between Dual Spacetime: A Biquaternionic Reformulation of General Relativity" (2025). `https://github.com/hypernumbernet/dual-spacetime-doc/blob/main/double-spacetime-theory.pdf`.

[3] Anonymous, "De Broglie Waves as Dual-Rotor Phase Lag: Quantum Mechanics as Geometry of Particle-Intrinsic Dual Spacetime" (2025). `https://github.com/hypernumbernet/dual-spacetime-doc/blob/main/dst-de-broglie-wave.pdf`.

[4] Anonymous, "Dual-Rotor Dynamics in Double Spacetime Theory: Written Action, Free Mismatch, and a Neighbouring Oscillator" (2025). `https://github.com/hypernumbernet/dual-spacetime-doc/blob/main/dst-dynamic-equation.pdf`.

[5] Anonymous, "Discrete Dual Spacetime Model and the Diophantine Structure of Mass Spectra" (2025). `https://github.com/hypernumbernet/dual-spacetime-doc/blob/main/discrete-dual-spacetime.pdf`.

[6] A. Einstein, "Riemann-Geometrie mit Aufrechterhaltung des Begriffes des Fernparallelismus," *Sitzungsber. Preuss. Akad. Wiss. Phys.-Math. Kl.*, 217–221 (1928).

[7] J. W. Maluf, "The teleparallel equivalent of general relativity," *Ann. Phys. (Berlin)* **525**, 339–359 (2013). doi:10.1002/andp.201200272

[8] R. Aldrovandi and J. G. Pereira, *Teleparallel Gravity: An Introduction*, Fundamental Theories of Physics **173**, Springer, Dordrecht (2013).

[9] S. Bahamonde, K. F. Dialektopoulos, C. Escamilla-Rivera, G. Farrugia, V. Gakis, M. Hendry, M. Hohmann, J. Levi Said, J. Mifsud and E. Di Valentino, "Teleparallel gravity: from theory to cosmology," *Rep. Prog. Phys.* **86**, 026901 (2023). arXiv:2106.13757 [gr-qc].

[10] P. R. Girard, "Einstein's equations and Clifford algebra," *Advances in Applied Clifford Algebras*, **9**, no. 2, 225–230 (1999). doi:10.1007/BF03042377

[11] D. Hestenes, "Spacetime physics with geometric algebra," *Am. J. Phys.* **71**, 691–714 (2003).

[12] C. Doran and A. Lasenby, *Geometric Algebra for Physicists*, Cambridge University Press, Cambridge (2003).

[13] P. Lounesto, *Clifford Algebras and Spinors*, 2nd ed., London Mathematical Society Lecture Note Series **286**, Cambridge University Press, Cambridge (2001).

[14] D. Hestenes, "Real Spinor Fields," *J. Math. Phys.* **8**, 798–808 (1967).

[15] P. A. M. Dirac, "The quantum theory of the electron," *Proc. R. Soc. Lond. A* **117**, 610–624 (1928).

[16] T. D. Newton and E. P. Wigner, "Localized states for elementary systems," *Rev. Mod. Phys.* **21**, 400–406 (1949).

[17] E. C. G. Stueckelberg, "Remarque à propos de la création de paires de particules en théorie de relativité," *Helv. Phys. Acta* **14**, 588–594 (1941).

[18] R. P. Feynman, "The theory of positrons," *Phys. Rev.* **76**, 749–759 (1949).

[19] P. W. Higgs, "Broken symmetries and the masses of gauge bosons," *Phys. Rev. Lett.* **13**, 508–509 (1964).

[20] F. Englert and R. Brout, "Broken symmetry and the mass of gauge vector mesons," *Phys. Rev. Lett.* **13**, 321–323 (1964).

[21] C. Rovelli, *Quantum Gravity*, Cambridge University Press, Cambridge (2004).

[22] A. Ashtekar and J. Lewandowski, "Background independent quantum gravity: a status report," *Class. Quantum Grav.* **21**, R53–R152 (2004).

[23] A. Ashtekar, "New variables for classical and quantum gravity," *Phys. Rev. Lett.* **57**, 2244–2247 (1986).

[24] B. S. DeWitt, "Quantum theory of gravity. I. The canonical theory," *Phys. Rev.* **160**, 1113–1148 (1967).

[25] J. Polchinski, *String Theory*, Vols. I and II, Cambridge University Press, Cambridge (1998).

[26] M. B. Green, J. H. Schwarz and E. Witten, *Superstring Theory*, Vols. 1 and 2, Cambridge University Press, Cambridge (1987).

[27] L. Bombelli, J. Lee, D. Meyer and R. D. Sorkin, "Space-time as a causal set," *Phys. Rev. Lett.* **59**, 521–524 (1987).

[28] J. Ambjørn, J. Jurkiewicz and R. Loll, "Reconstructing the universe," *Phys. Rev. D* **72**, 064014 (2005).

[29] M. Niedermaier and M. Reuter, "The asymptotic safety scenario in quantum gravity," *Living Rev. Relativ.* **9**, 5 (2006).

[30] T. Jacobson, "Thermodynamics of spacetime: the Einstein equation of state," *Phys. Rev. Lett.* **75**, 1260–1263 (1995).

[31] E. P. Verlinde, "On the origin of gravity and the laws of Newton," *J. High Energy Phys.* **04**, 029 (2011).

[32] S. W. Hawking, "Particle creation by black holes," *Commun. Math. Phys.* **43**, 199–220 (1975).

[33] A. Almheiri, D. Marolf, J. Polchinski and J. Sully, "Black holes: complementarity or firewalls?" *J. High Energy Phys.* **02**, 062 (2013).

[34] V. Cardoso and P. Pani, "Testing the nature of dark compact objects: a status report," *Living Rev. Relativ.* **22**, 4 (2019).

[35] J. Abedi, H. Dykaar and N. Afshordi, "Echoes from the Abyss: Tentative evidence for Planck-scale structure at black hole horizons," *Phys. Rev. D* **96**, 082004 (2017).

[36] The Event Horizon Telescope Collaboration, "First M87 Event Horizon Telescope results. I. The shadow of the supermassive black hole," *Astrophys. J. Lett.* **875**, L1 (2019).

[37] The Event Horizon Telescope Collaboration, "First Sagittarius A* Event Horizon Telescope results. I. The shadow of the supermassive black hole in the center of the Milky Way," *Astrophys. J. Lett.* **930**, L12 (2022).

[38] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), "Observation of gravitational waves from a binary black hole merger," *Phys. Rev. Lett.* **116**, 061102 (2016).

[39] P. Amaro-Seoane *et al.*, "Laser Interferometer Space Antenna," arXiv:1702.00786 [astro-ph.IM] (2017).

[40] M. Punturo *et al.*, "The Einstein Telescope: a third-generation gravitational wave observatory," *Class. Quantum Grav.* **27**, 194002 (2010).

[41] A. H. Guth, "Inflationary universe: a possible solution to the horizon and flatness problems," *Phys. Rev. D* **23**, 347–356 (1981).

[42] A. A. Starobinsky, "A new type of isotropic cosmological models without singularity," *Phys. Lett. B* **91**, 99–102 (1980).

[43] S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1–23 (1989).

[44] J. S. Bell, "On the Einstein Podolsky Rosen paradox," *Physics Physique Fizika* **1**, 195–200 (1964).

[45] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, "Proposed experiment to test local hidden-variable theories," *Phys. Rev. Lett.* **23**, 880–884 (1969).

[46] A. Einstein, B. Podolsky and N. Rosen, "Can quantum-mechanical description of physical reality be considered complete?" *Phys. Rev.* **47**, 777–780 (1935).